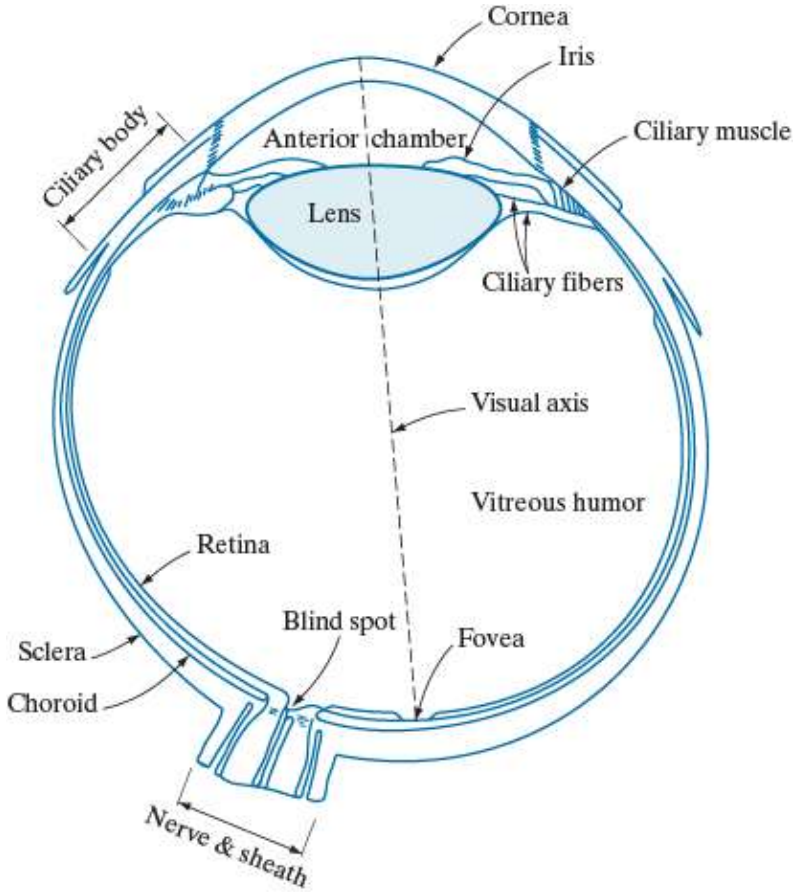
Image Processing – Ch2



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Human Visual Perception

- **Human intuition and analysis play a central role in the choice of DIP techniques**
  - Physical limitations of human vision

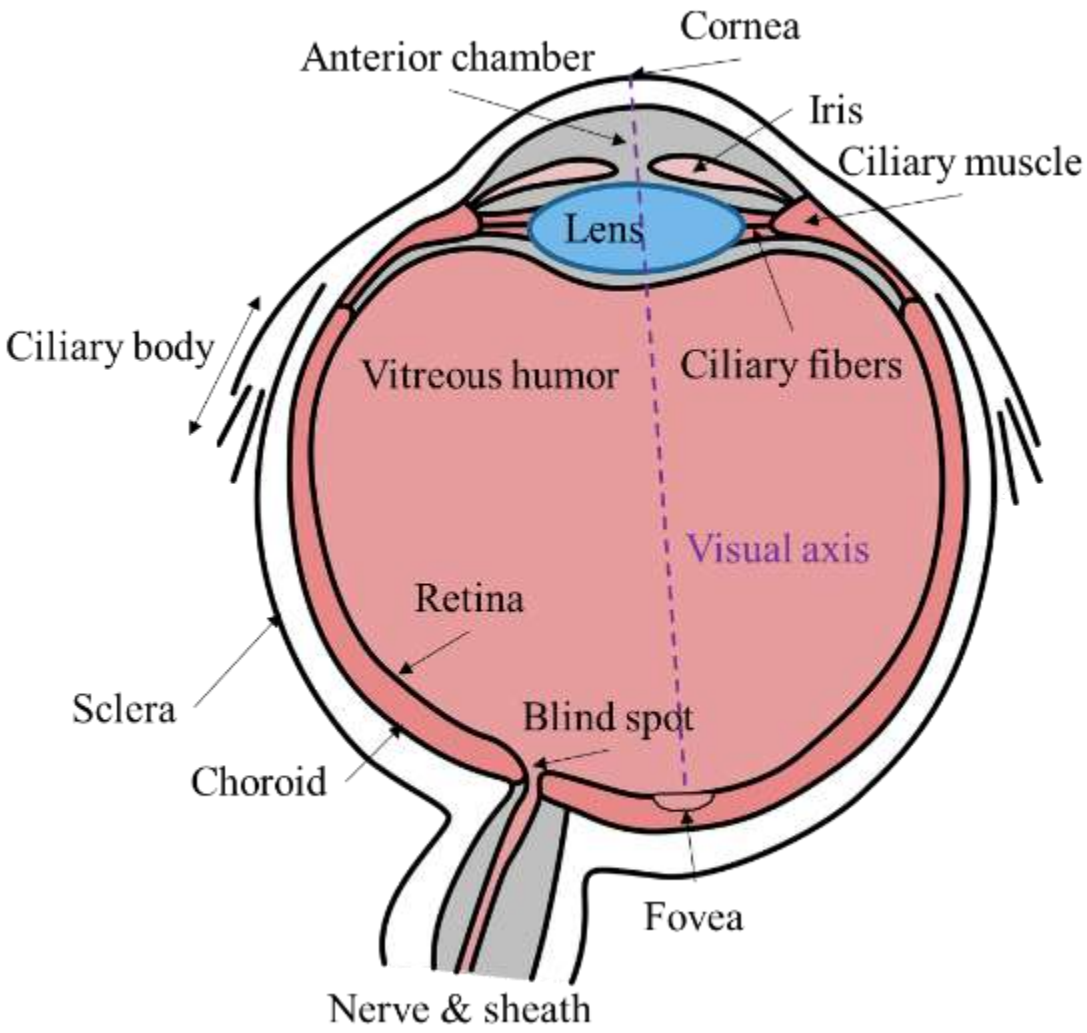

FIGURE 2.1 Simplified diagram of a cross section of the human eye.

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Structure of the Human Eye

- **Pattern vision is afforded by the distribution of discrete light receptors over the surface of the retina.**
- **There are two classes of receptors: cones and rods.**
  - The number of cones in each eye: 6 to 7 millions
  - The number of rods in each eye: 75 to 150 millions
  - The cones is concentrated in the central portion of the retina (fovea).
  - The rods are distributed over the retinal surface.
- **Photopic vision (bright-light): vision with cones**
  - color receptors, high resolution in the fovea, less sensitive to light
- **Scotopic vision (dim-light) : vision with rods**
  - color blind, much more sensitive to light (night vision), lower resolution

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Distribution of Rods and Cones

- The number of cones in each eye: 6 to 7 millions
- The number of rods in each eye: 75 to 150 millions

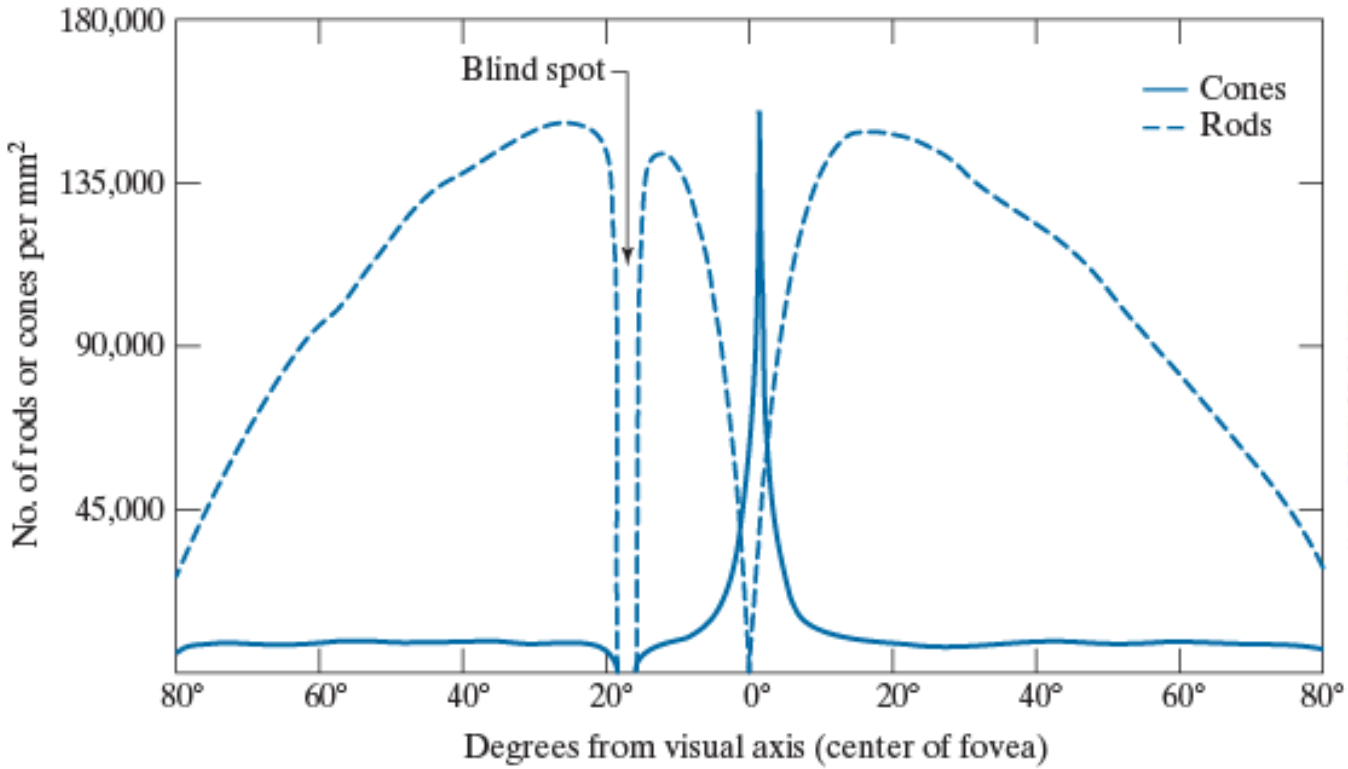

FIGURE 2.2 Distribution of rods and cones in the retina.

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Image Formation in the Eye

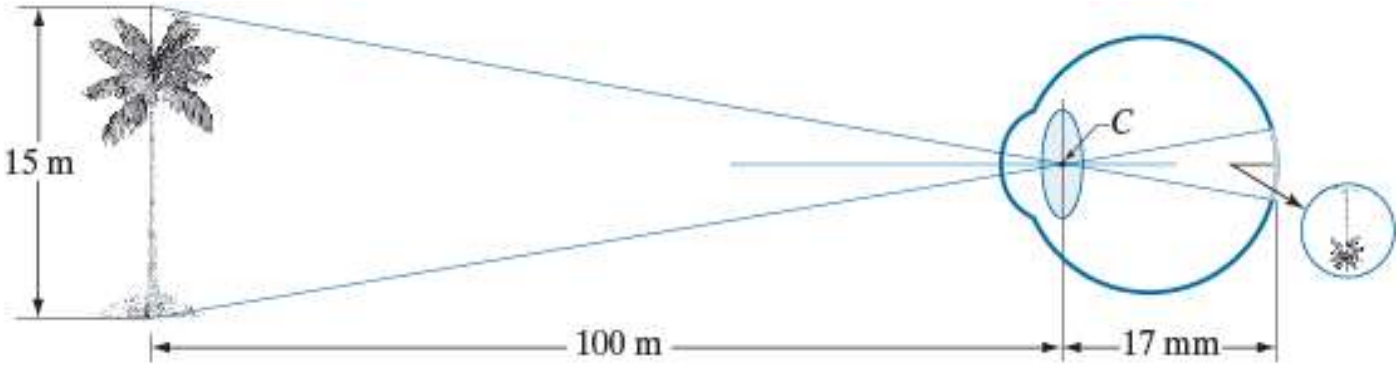

FIGURE 2.3 Graphical representation of the eye looking at a palm tree. Point *C* is the focal center of the lens.

- **Focal length of the eye: 17 to 14 mm**
- **Let *h* be the height in mm of that object in the retinal image, then**

$$15/100 = h / 17 \text{ , } h = 2.55\text{mm}$$

- **The retinal image is reflected primarily in the area of the fovea**
- **Perception takes place by the relative excitation of light receptors which transform radiant energy into electrical impulses**

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Brightness Adaptation and Discrimination

- **The range of brightness that the eye can adapt to is enormous**
- **Photopic vision alone has a range of around 10^6 However, not over such range simultaneously**
- **Brightness adaptation**
  - Brightness adaptation level, B_a . B_b being the threshold that all intensity levels below is indistinguishable blacks
- **mL: millilambert**

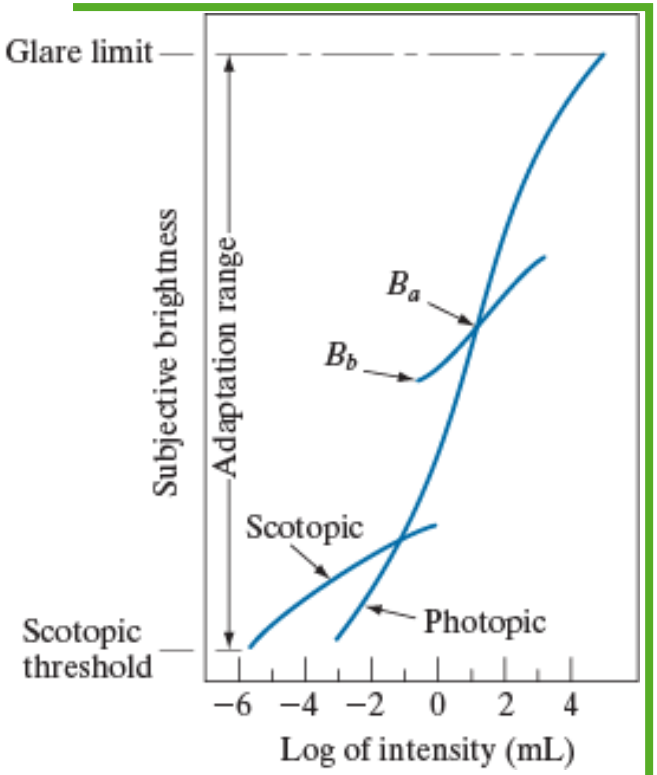

FIGURE 2.4 Range of subjective brightness sensations showing a particular adaptation level, B_a.

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Weber Ratio

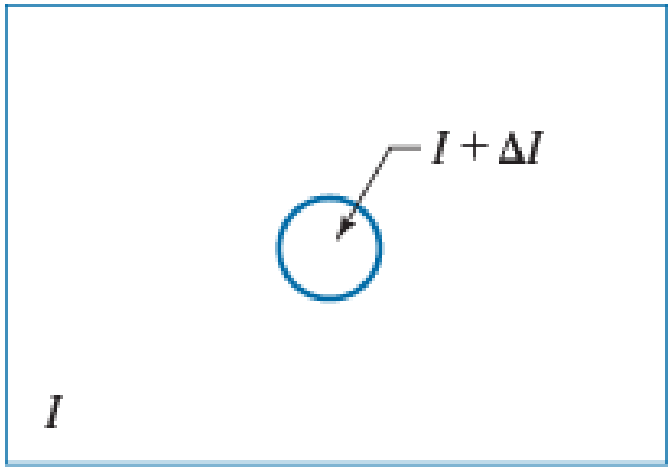

FIGURE 2.5 Basic experimental setup used to characterize brightness discrimination.

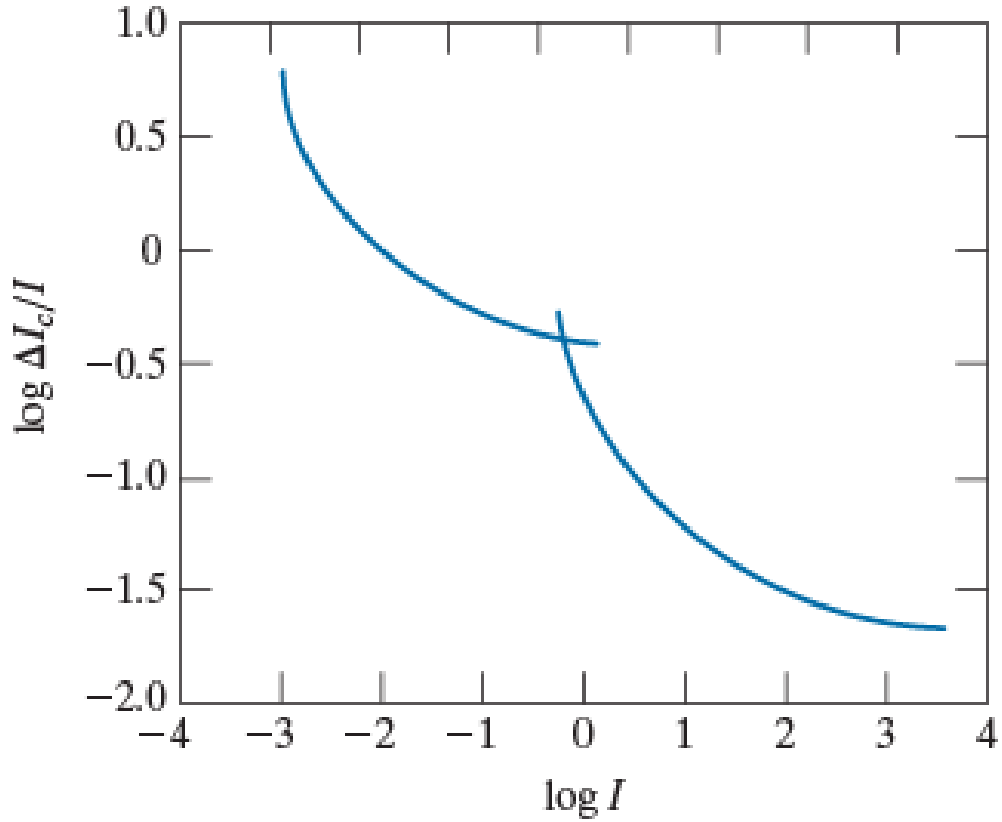

FIGURE 2.6 Typical Weber ratio as a function of intensity.

- **Weber ratio: $\Delta I_c/I$**
- **Discriminable 50% of the time**
- **Brightness discrimination is poor at low levels of illumination**

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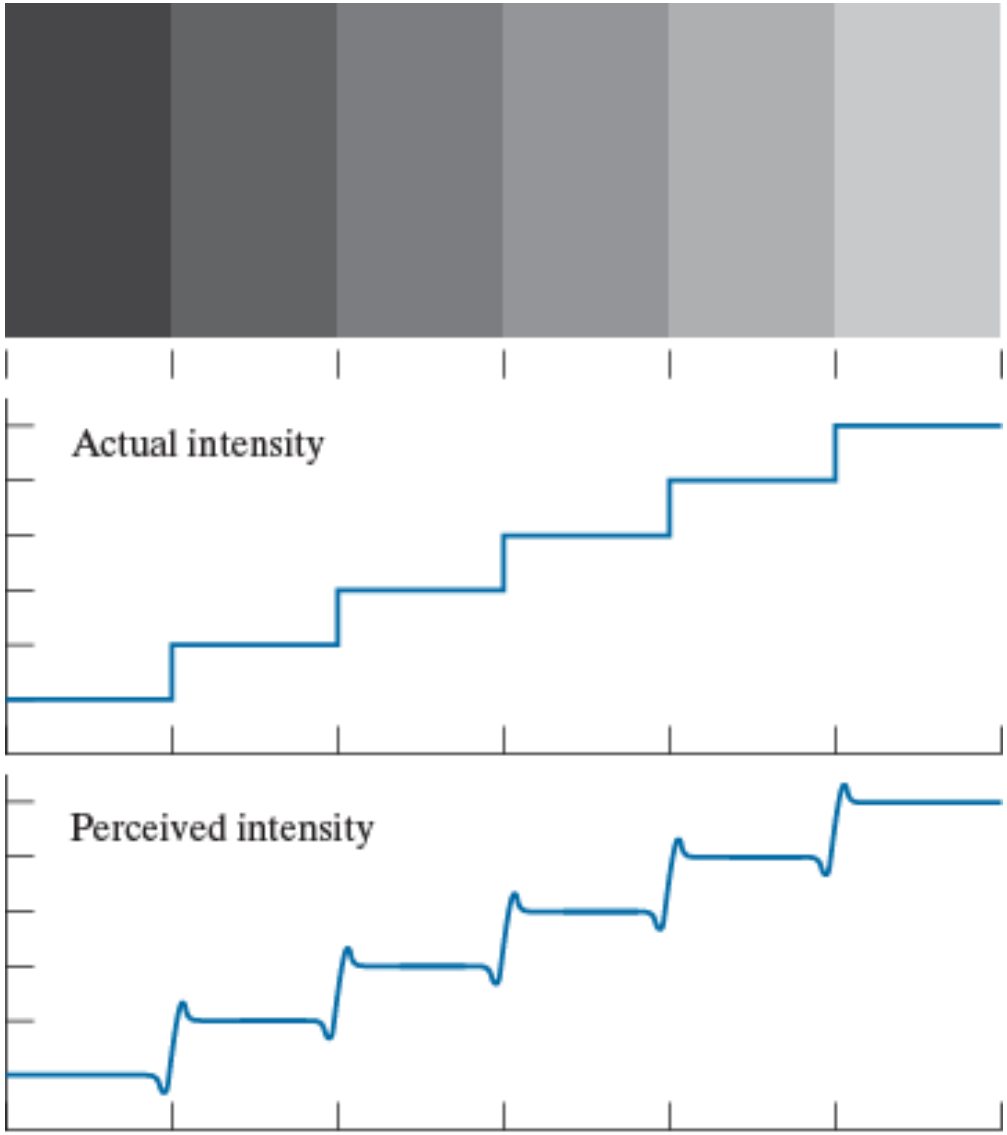
Mach Band Effect



a
b
c

FIGURE 2.7 Illustration of the Mach band effect. Perceived intensity is not a simple function of actual intensity.

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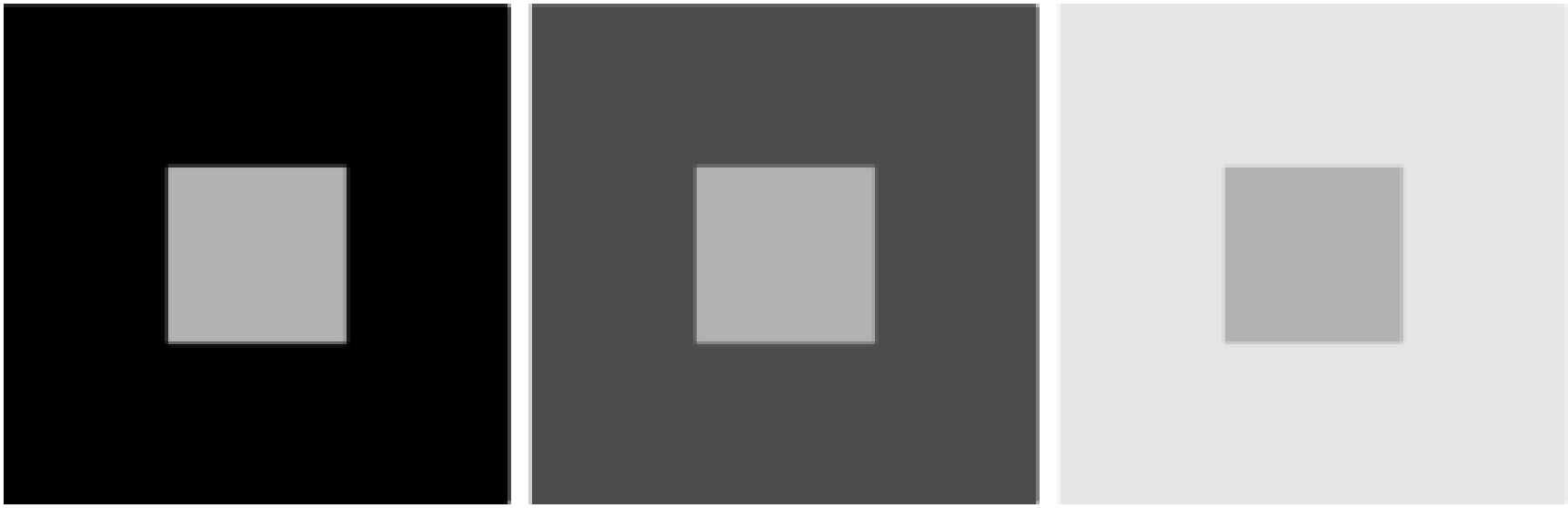
Simultaneous Contrast



a b c

FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

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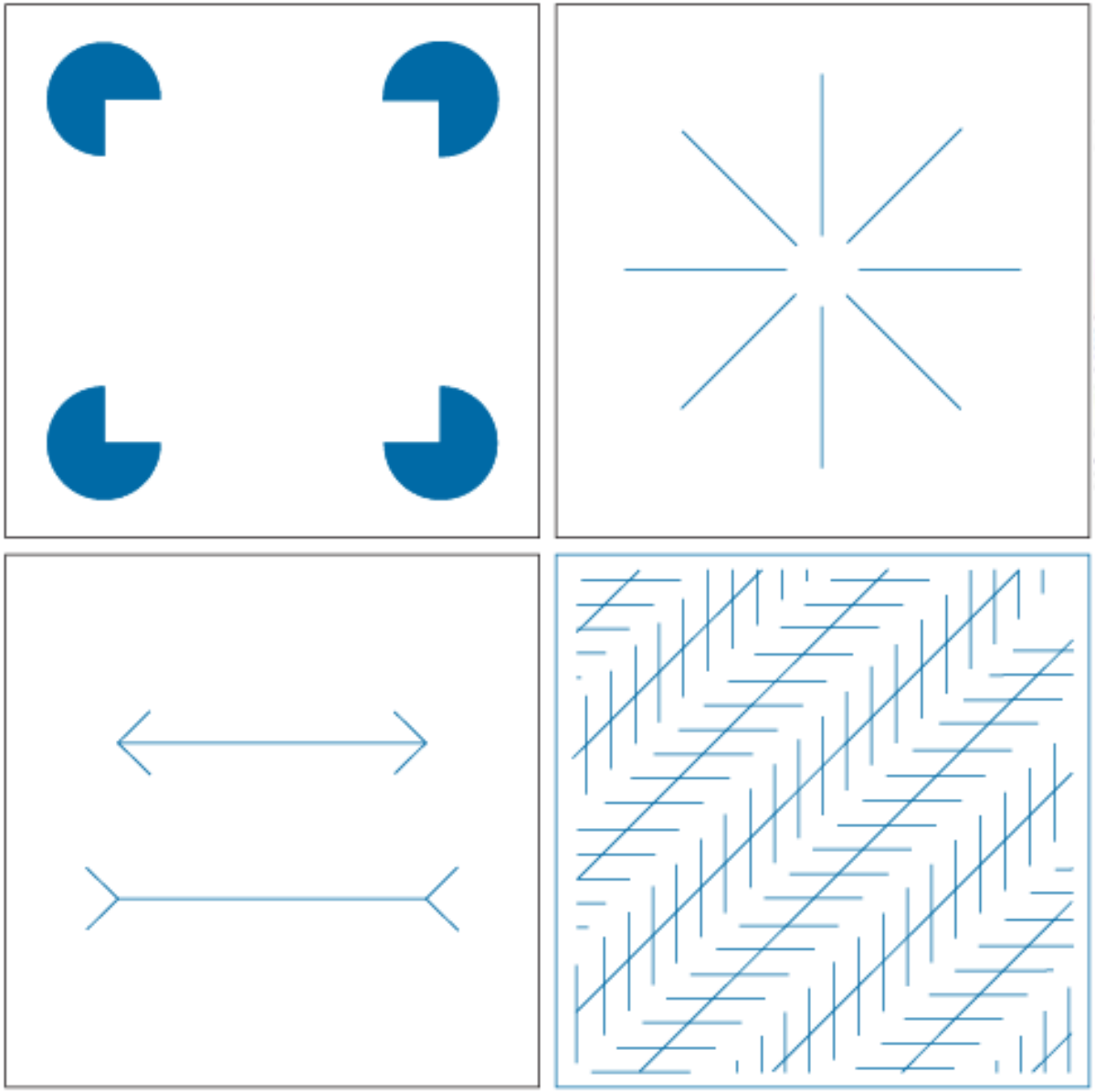
Optical Illusions



a b
c d

FIGURE 2.9 Some well-known optical illusions.

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Effect of Blind Spot

https://www.dailymail.co.uk/home/index.html

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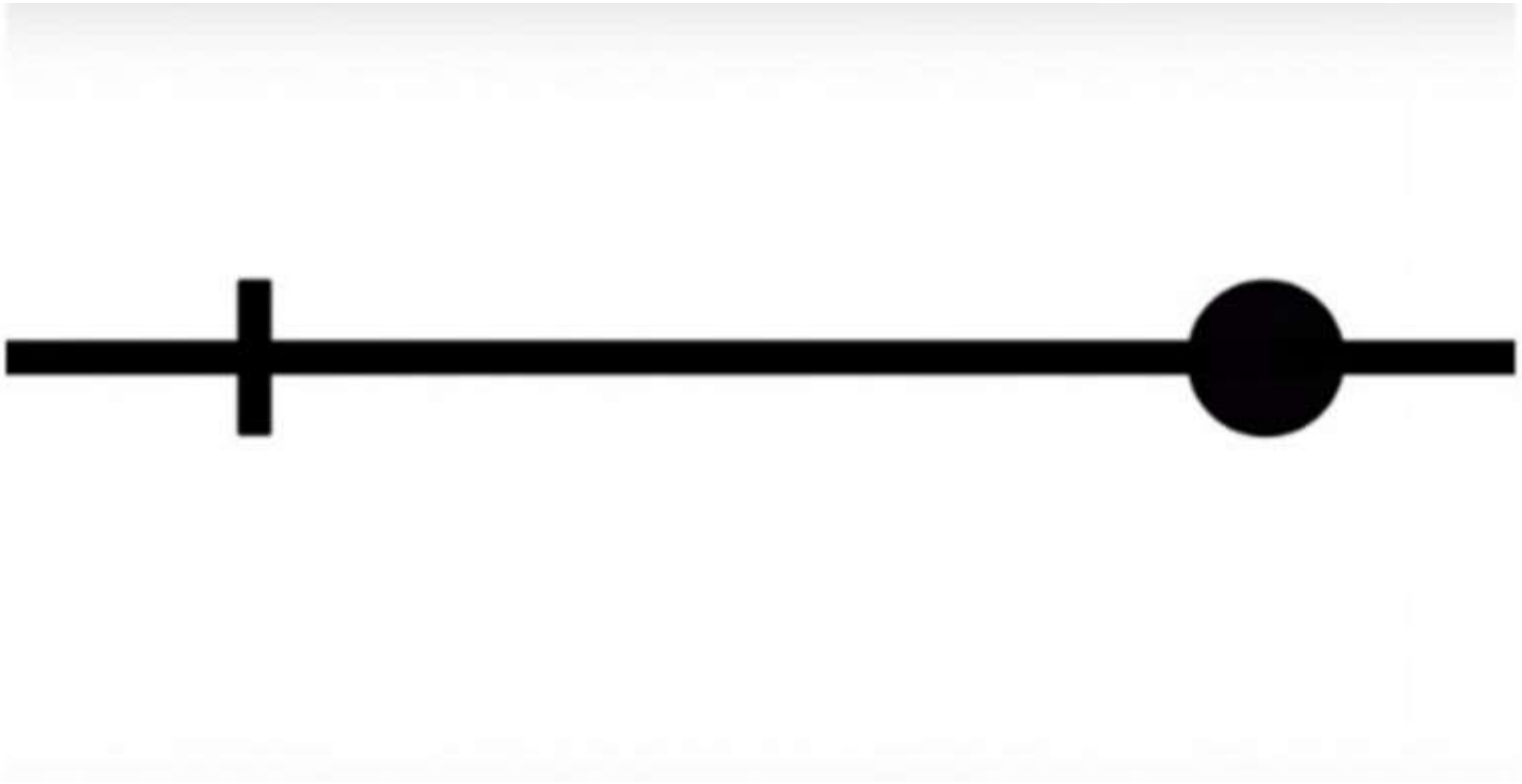
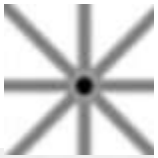
Effect of Blind Spot and Your Brain

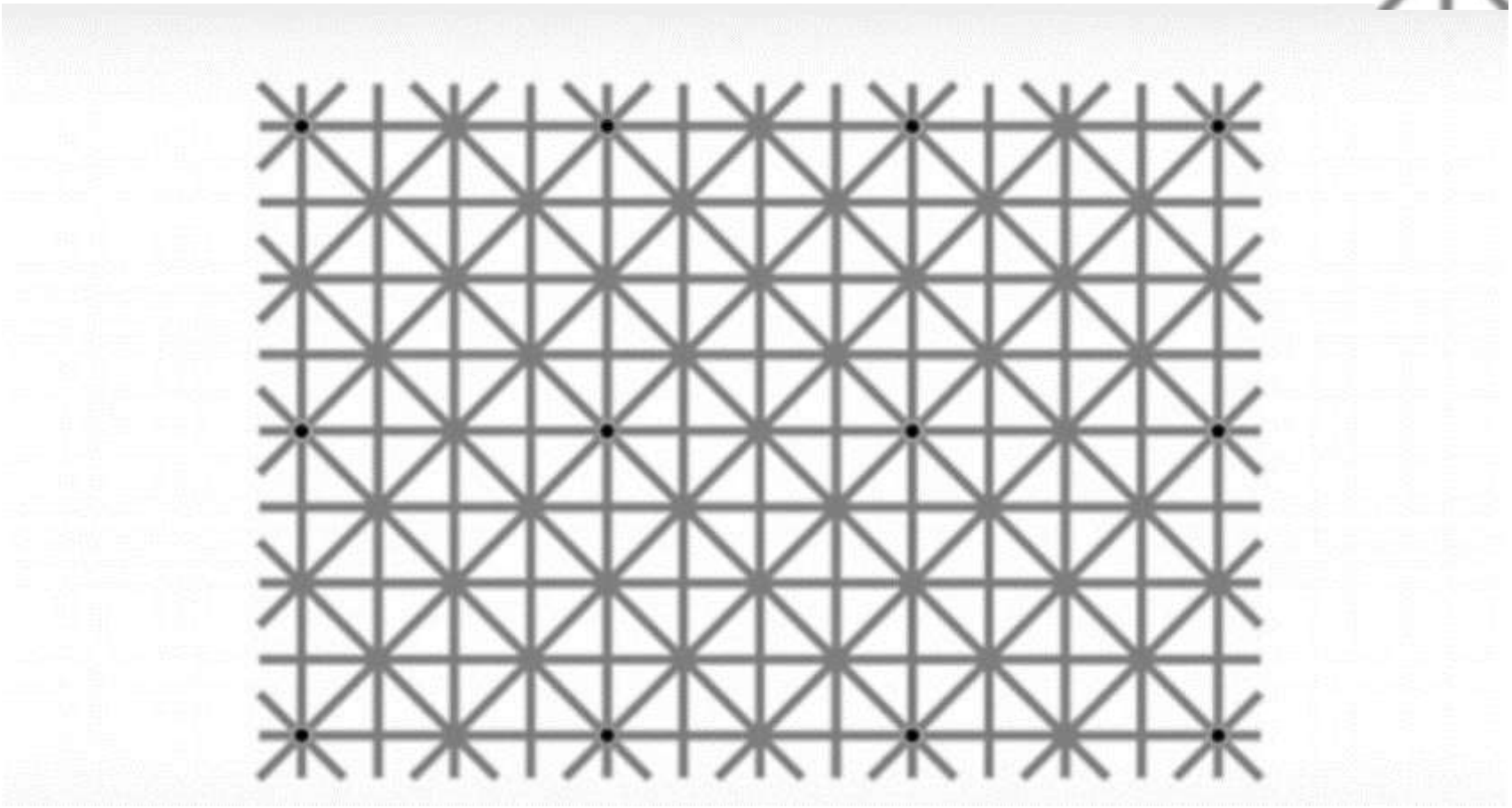


https://www.dailymail.co.uk/home/index.html

為保護所參考來源之智慧財產權，此教材僅為上課時教學之用。

What You See May not be What it is

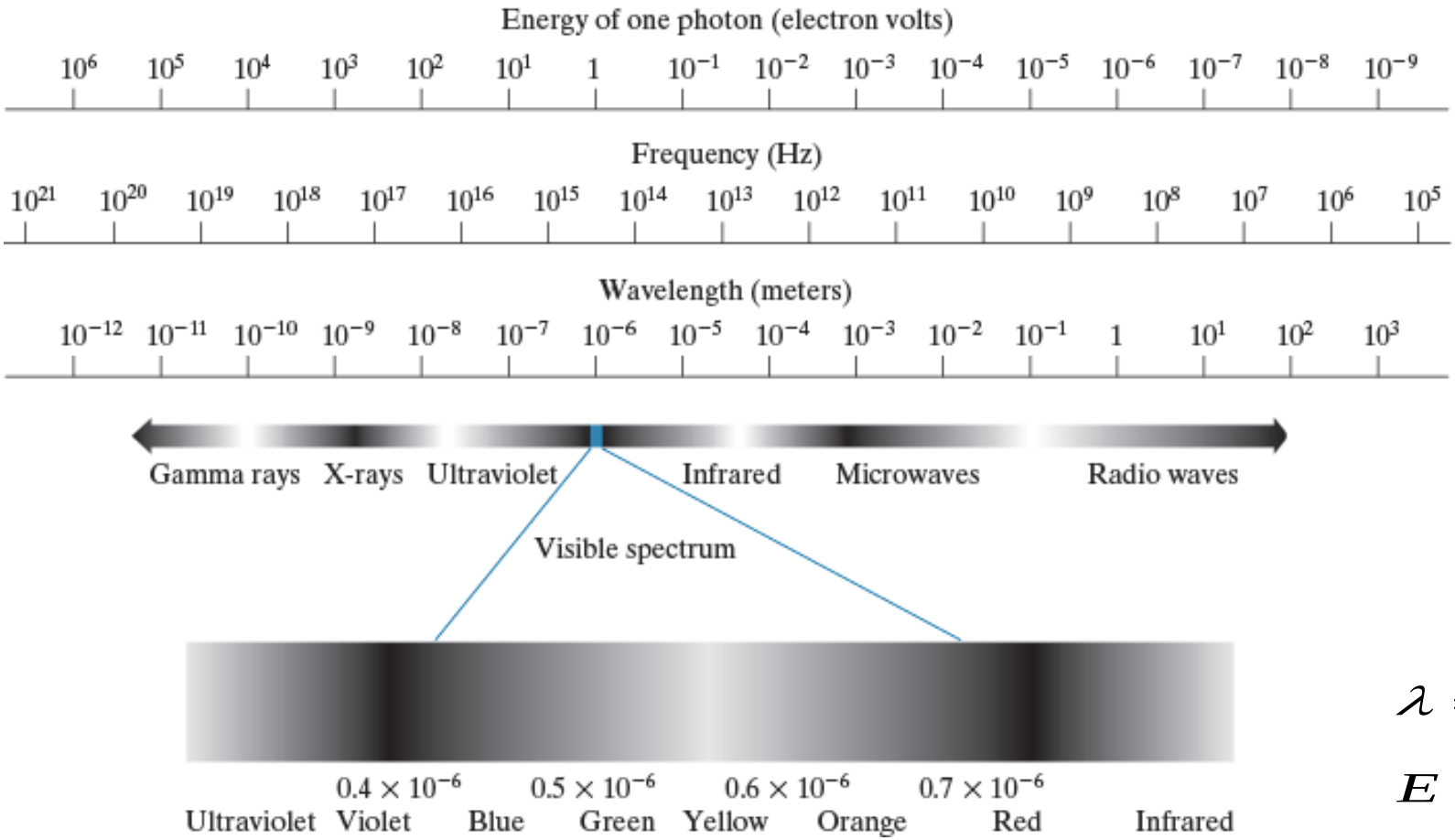




https://www.dailymail.co.uk/home/index.html

為保護所參考來源之智慧財產權，此教材僅為上課時教學之用。

Light and the EM Spectrum



$$\lambda = \frac{c}{\nu}$$

$$E = h\nu$$

FIGURE 2.10 The electromagnetic spectrum. The visible spectrum is shown zoomed to facilitate explanations, but note that it encompasses a very narrow range of the total EM spectrum.

為保護所參考來源之智慧財產權，此教材僅為上課時教學之用。

Light and the EM Spectrum

- **Relationship between frequency ν and wavelength** $\lambda = \frac{c}{\nu}$
- **Energy of a photon** $E = h\nu$ **, where *h* is Planck's constant**

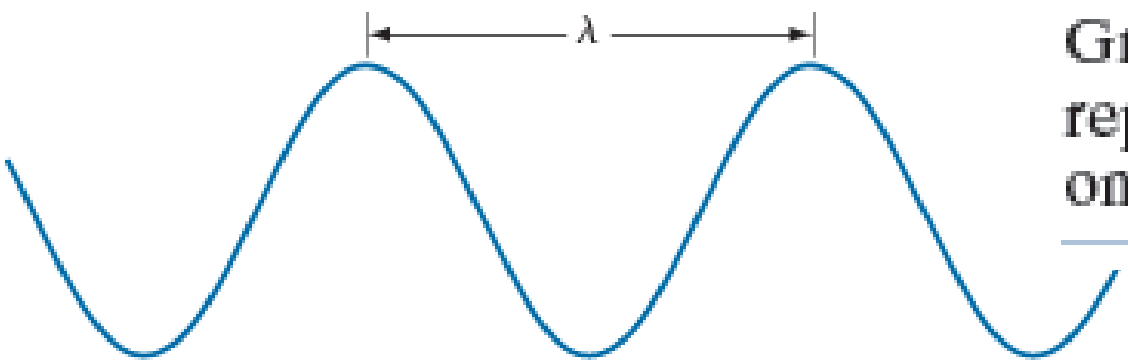


FIGURE 2.11 Graphical representation of one wavelength.

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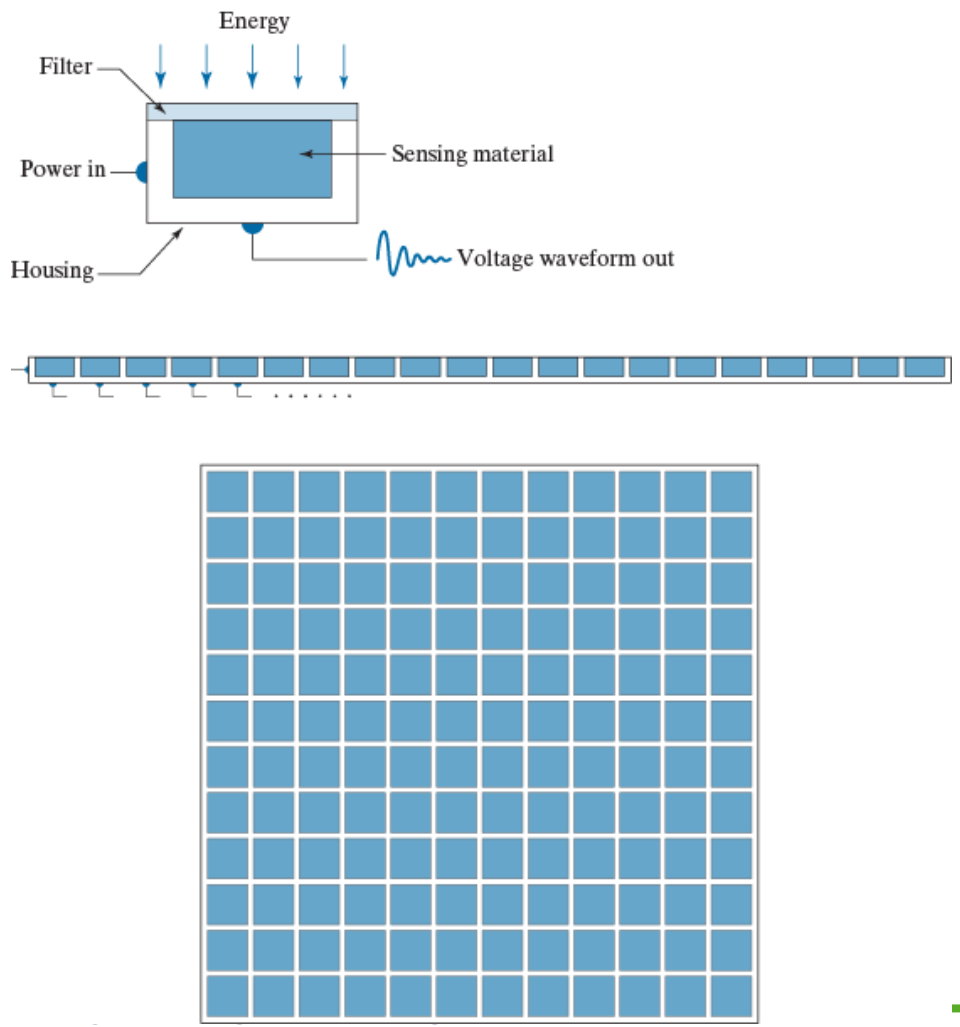
Image Sensing and Acquisition

- **Nowadays most visible and near IR electromagnetic imaging is done with 2-dimensional charged-coupled devices (CCDs).**



a
b
c

FIGURE 2.12
(a) Single sensing element.
(b) Line sensor.
(c) Array sensor.

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Image Sensing and Acquisition

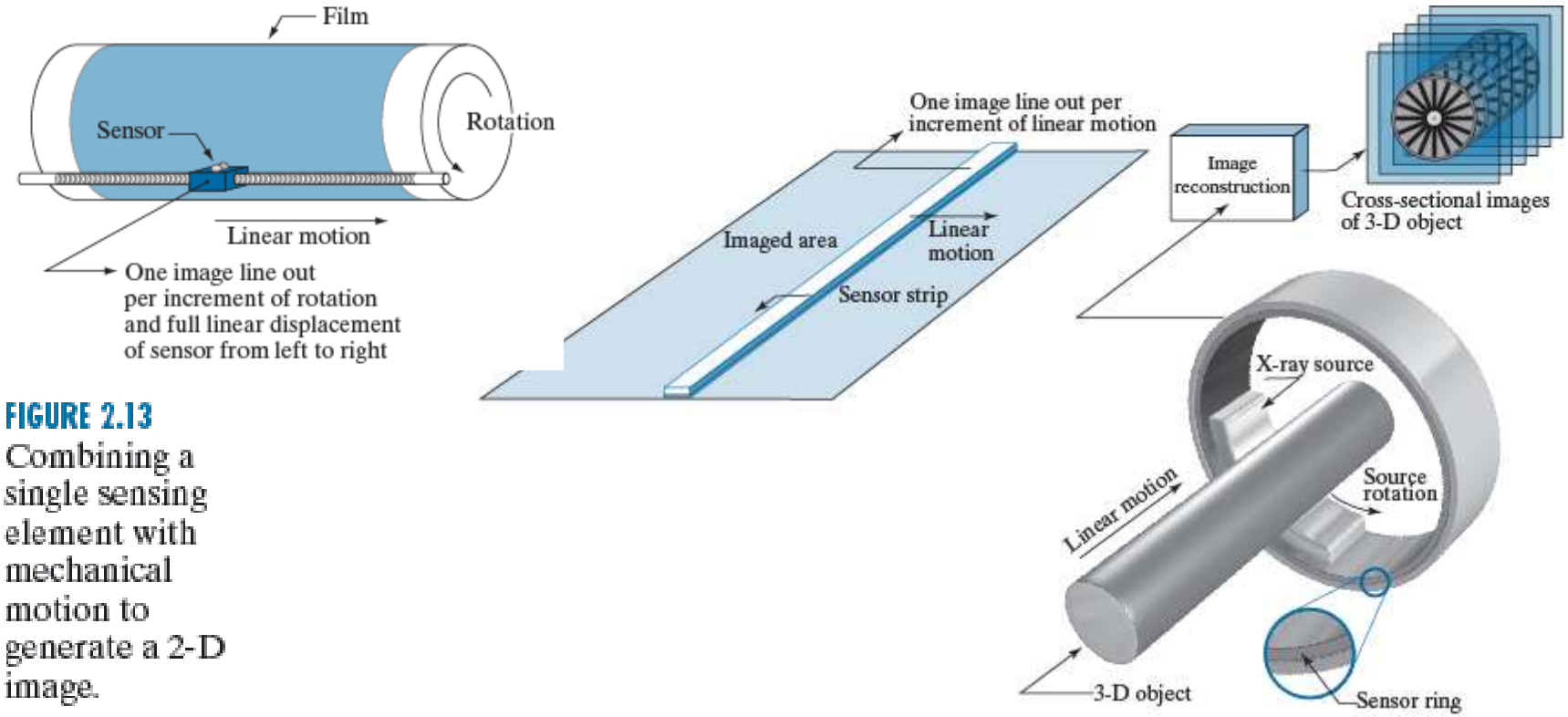

FIGURE 2.13 Combining a single sensing element with mechanical motion to generate a 2-D image.

Computerized Axial Tomography (CAT)
Magnetic Resonance Imaging (MRI)
Positron Emission Tomography (PET)

a b

FIGURE 2.14 (a) Image acquisition using a linear sensor strip. (b) Image acquisition using a circular sensor strip.

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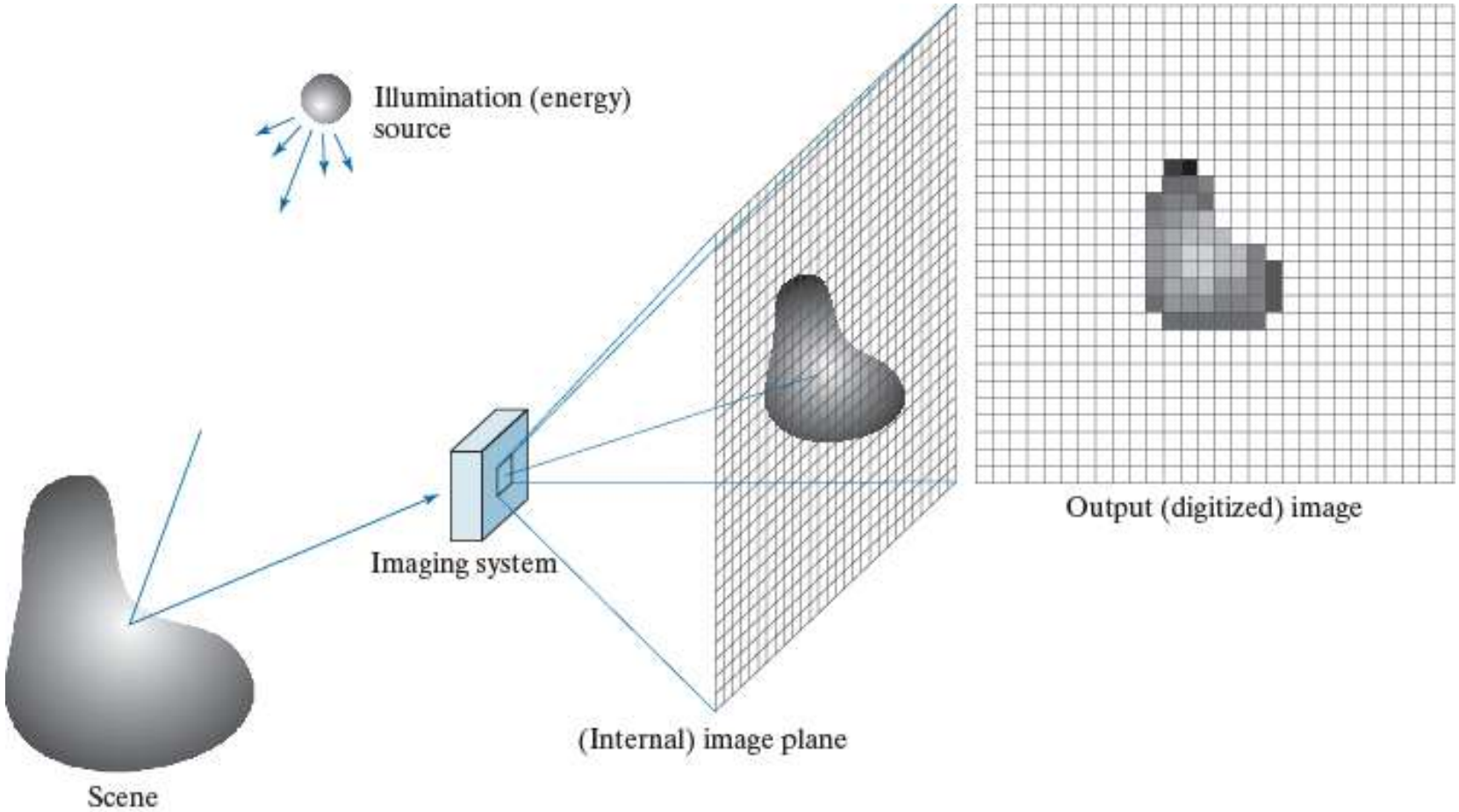
Image Sensing and Acquisition



a b c d e

FIGURE 2.15 An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

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Image Formation Model

- **Binary images:** images having only two possible brightness levels (black and white)
- **Gray scale images :** "black and white" images
- **Color images:** can be described mathematically as three gray scale images

- Let $f(x,y)$ be an image function, then

$$f(x,y) = i(x,y)\, r(x,y),$$

where $i(x,y)$: the illumination function

$r(x,y)$: the reflection function

Note: $0 < i(x,y) < \infty$ and $0 < r(x,y) < 1$.

- For digital images the minimum gray level is usually 0, but the maximum depends on number of quantization levels used to digitize an image. The most common is 256 levels, so that the maximum level is 255.

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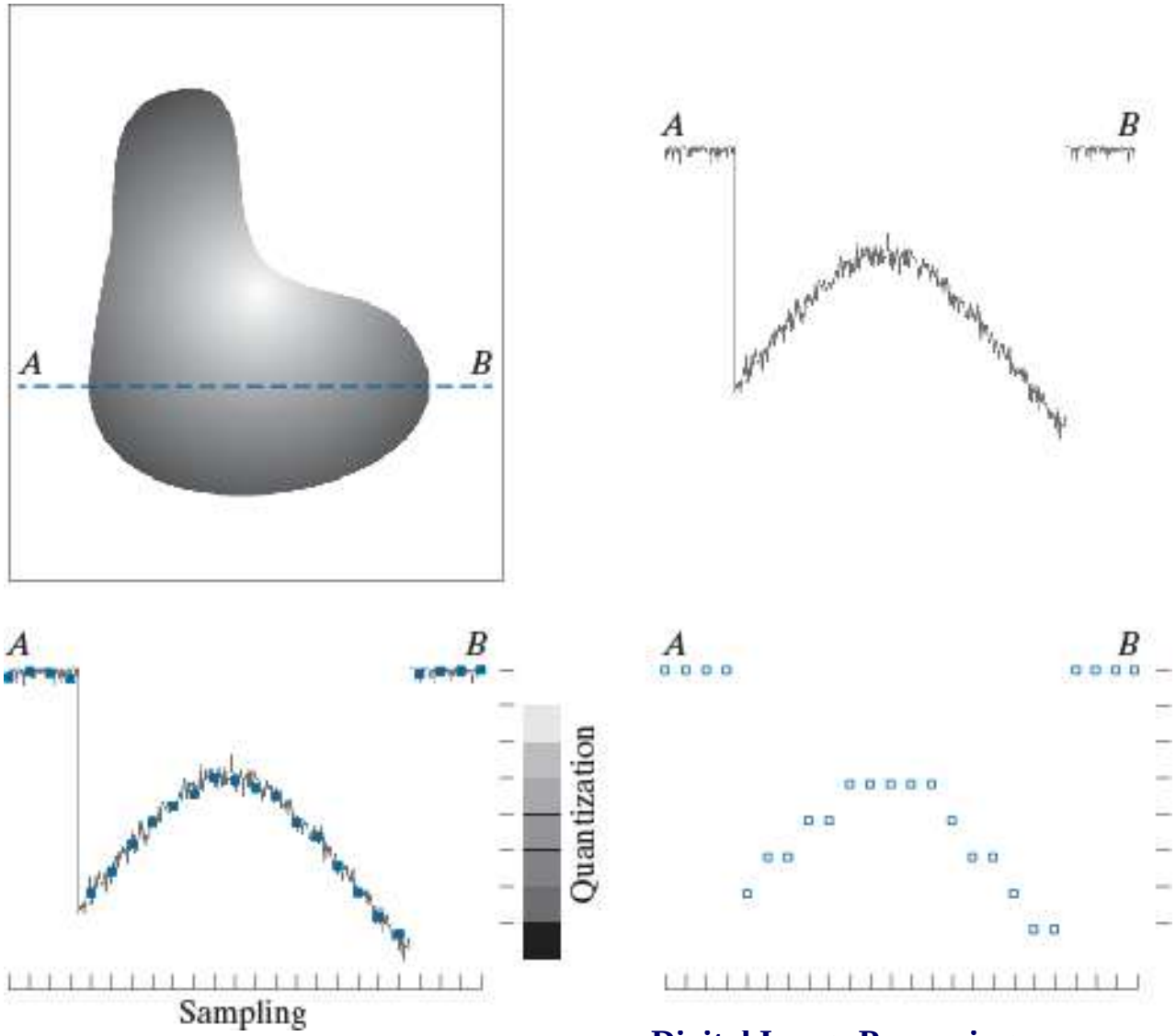
Image Sampling and Quantization



a b
c d

FIGURE 2.16 (a) Continuous image. (b) A scan line showing intensity variations along line *AB* in the continuous image. (c) Sampling and quantization. (d) Digital scan line. (The black border in (a) is included for clarity. It is not part of the image).

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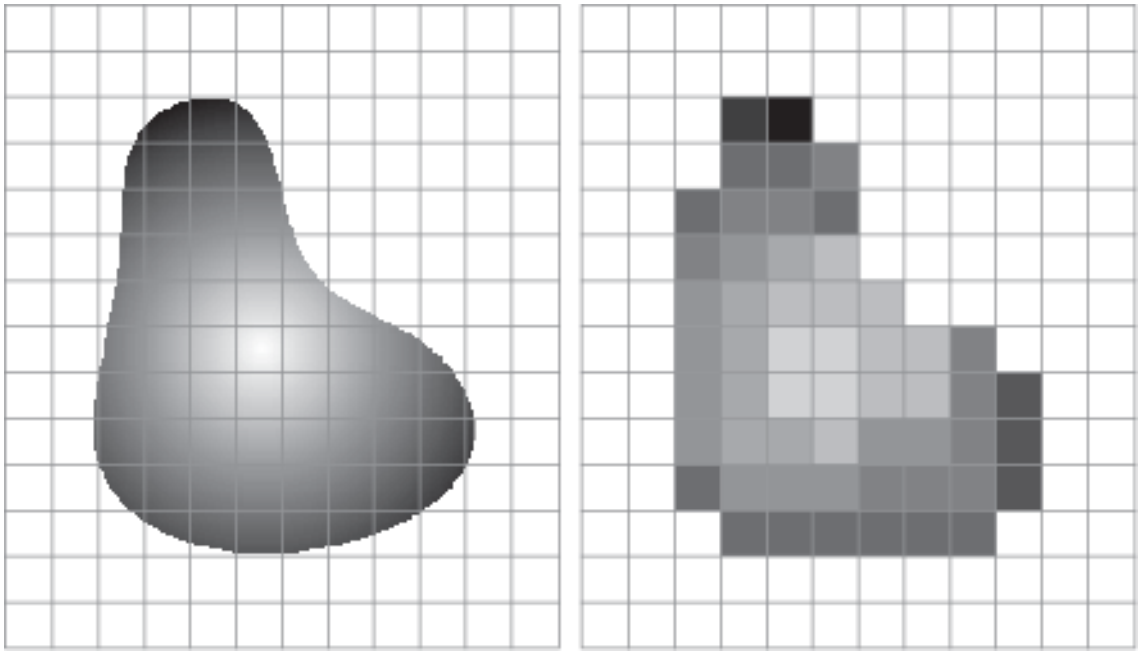
Image Sampling and Quantization

- **Sampling:** digitizing the 2-dimensional spatial coordinate values
- **Quantization:** digitizing the amplitude values (brightness level)



a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

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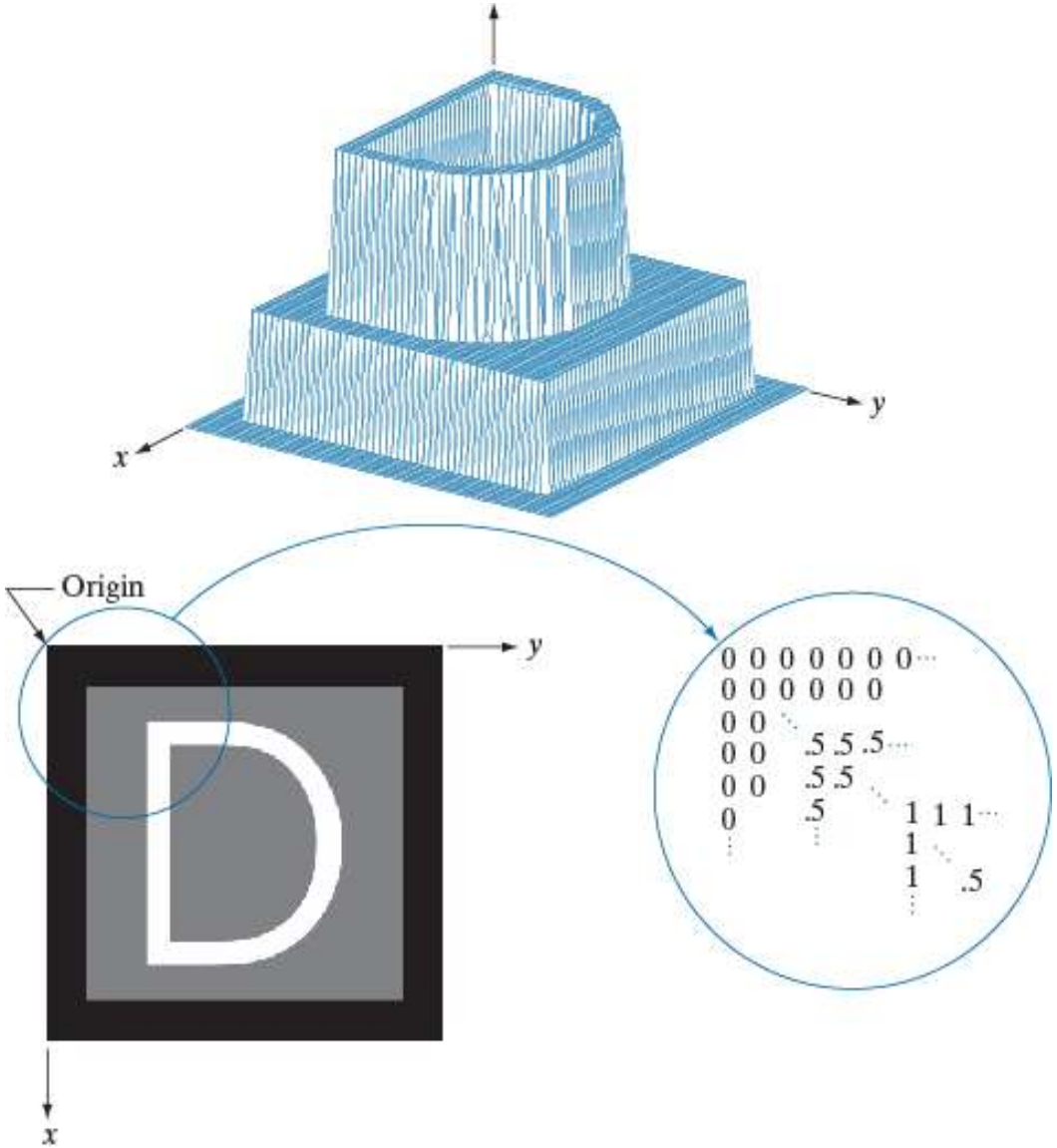
Representing Digital Images



a
b c

FIGURE 2.18 (a) Image plotted as a surface. (b) Image displayed as a visual intensity array. (c) Image shown as a 2-D numerical array. (The numbers 0, .5, and 1 represent black, gray, and white, respectively.)

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Saturation and Noise

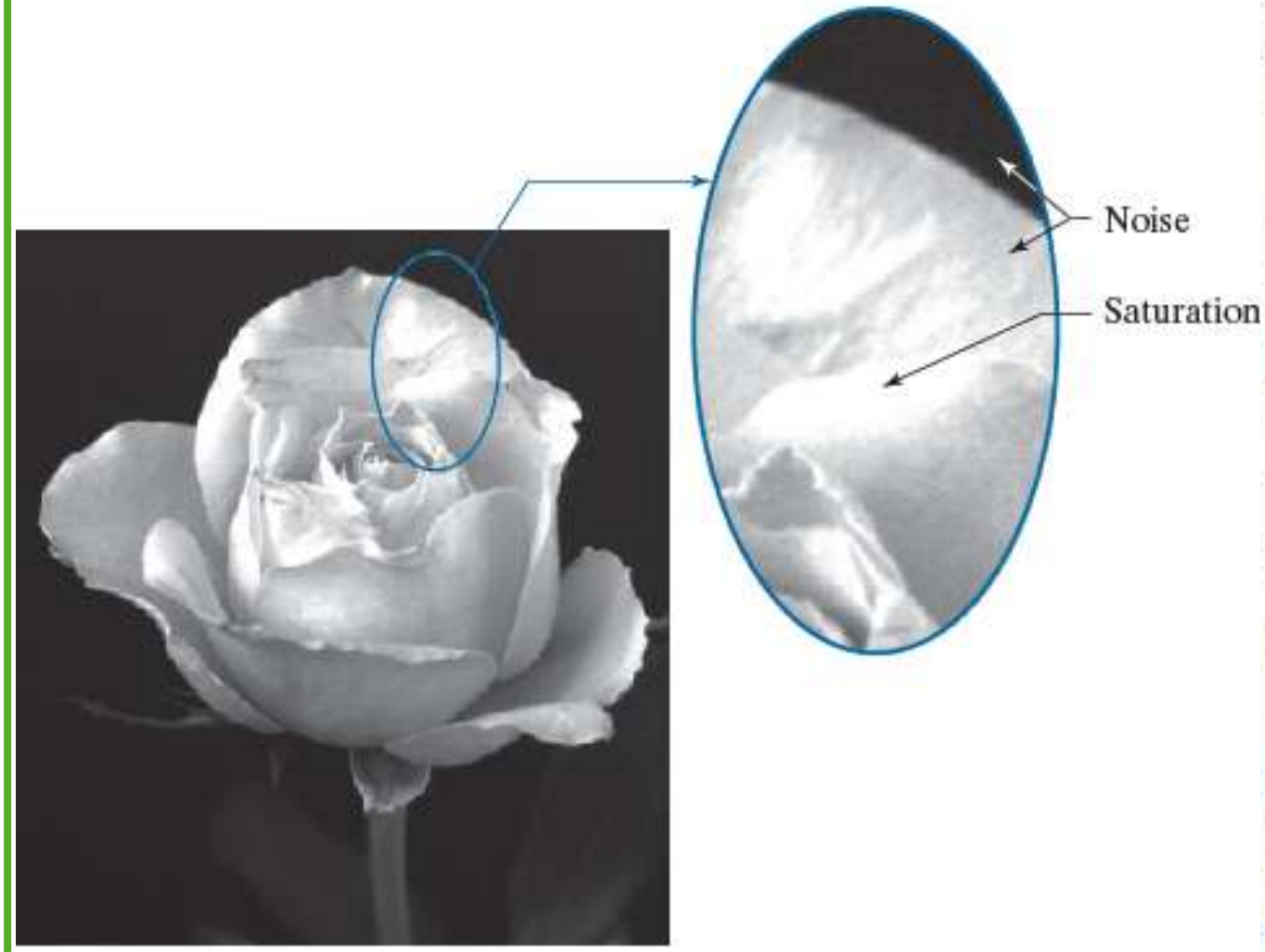

FIGURE 2.20 An image exhibiting saturation and noise. Saturation is the highest value beyond which all intensity values are clipped (note how the entire saturated area has a high, constant intensity level). Visible noise in this case appears as a grainy texture pattern. The dark background is noisier, but the noise is difficult to see.

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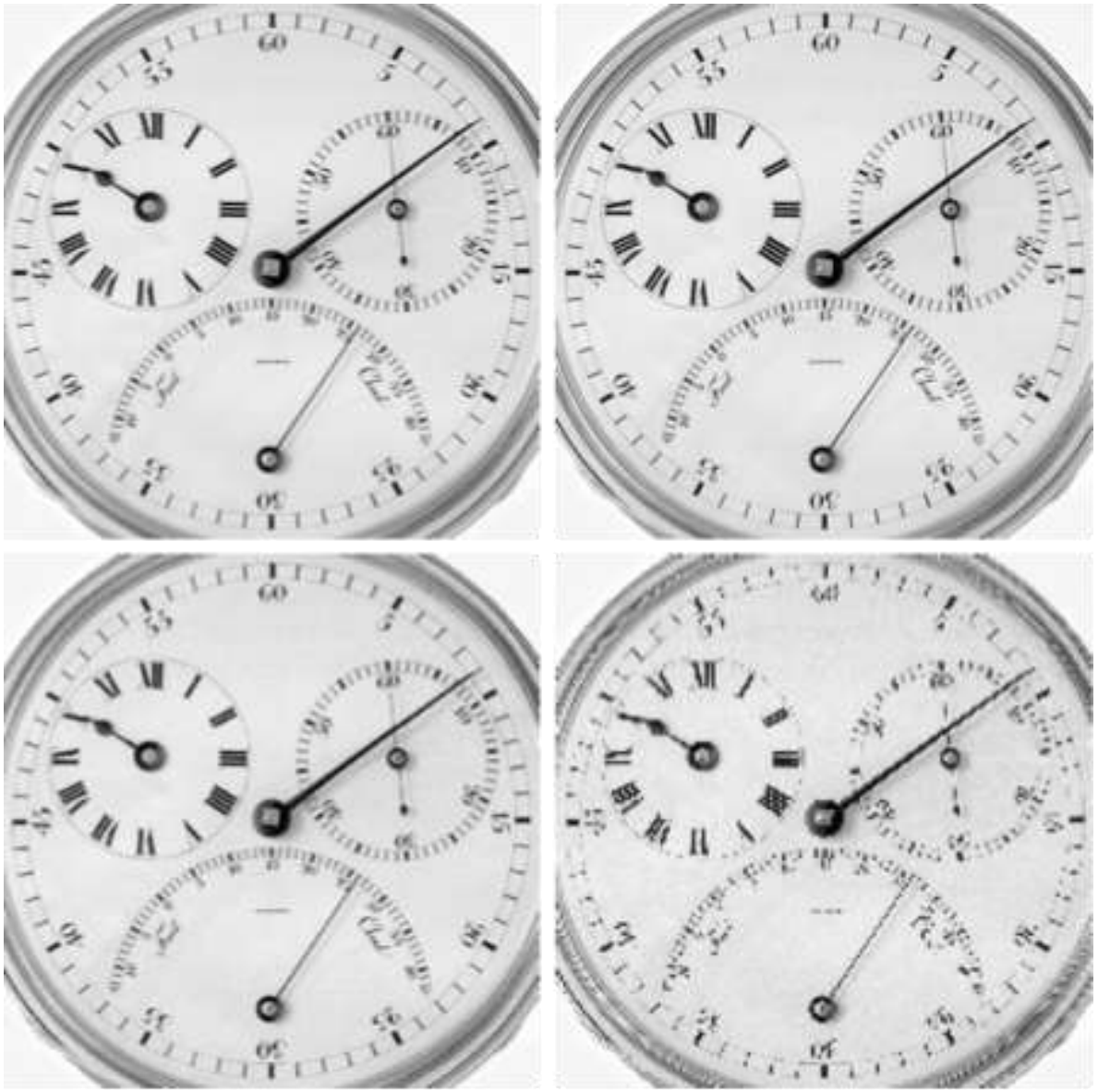
Image Resolution



a b
c d

FIGURE 2.23 Effects of reducing spatial resolution. The images shown are at: (a) 930 dpi, (b) 300 dpi, (c) 150 dpi, and (d) 72 dpi.

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Intensity Resolution

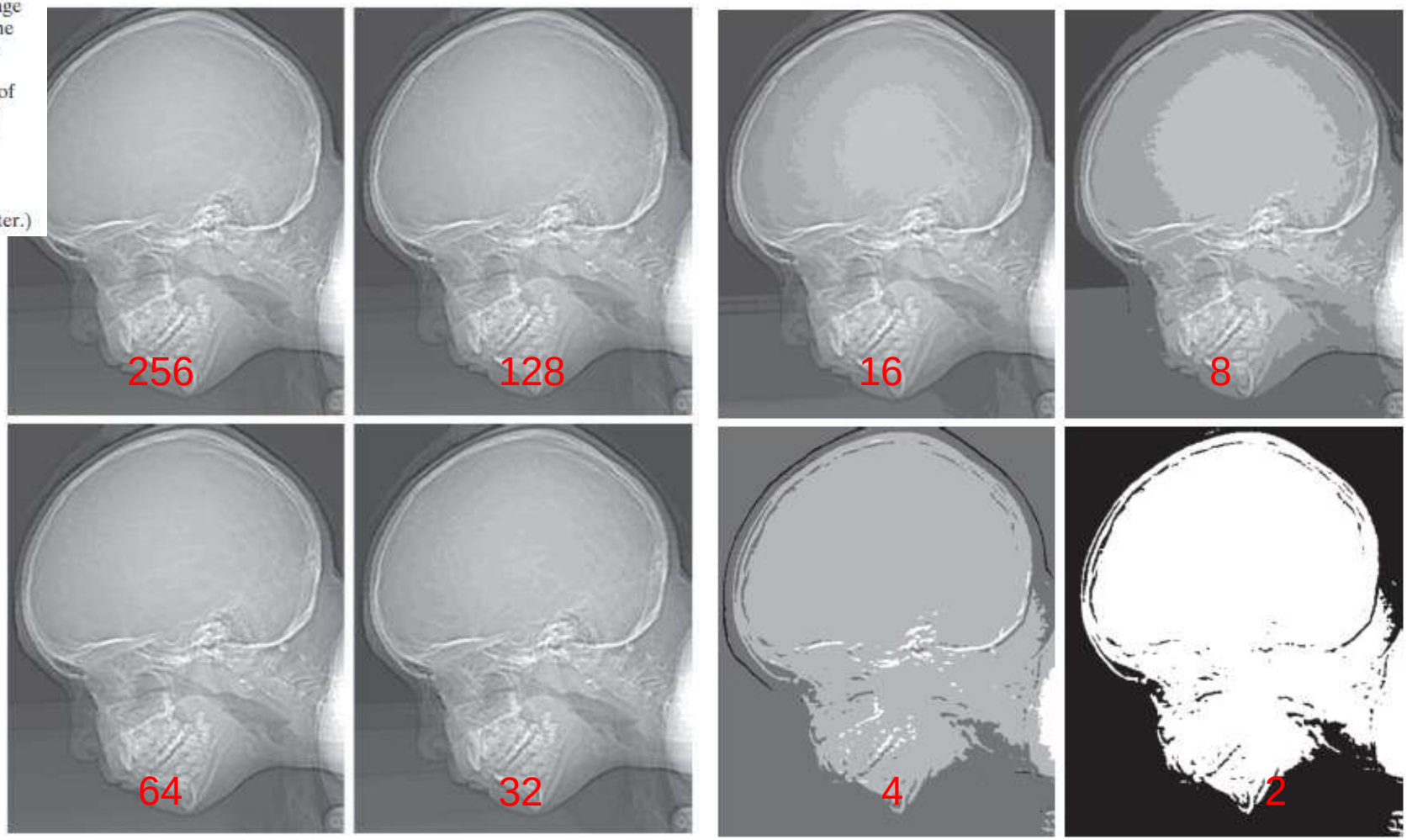
PS: false contouring

a b
c d

FIGURE 2.24 (a) 774 × 640, 256-level image. (b)-(d) Image displayed in 128, 64, and 32 intensity levels, while keeping the spatial resolution constant. (Original image courtesy of the Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)

e f
g h

FIGURE 2.24 (*Continued*) (e)-(h) Image displayed in 16, 8, 4, and 2 intensity levels.



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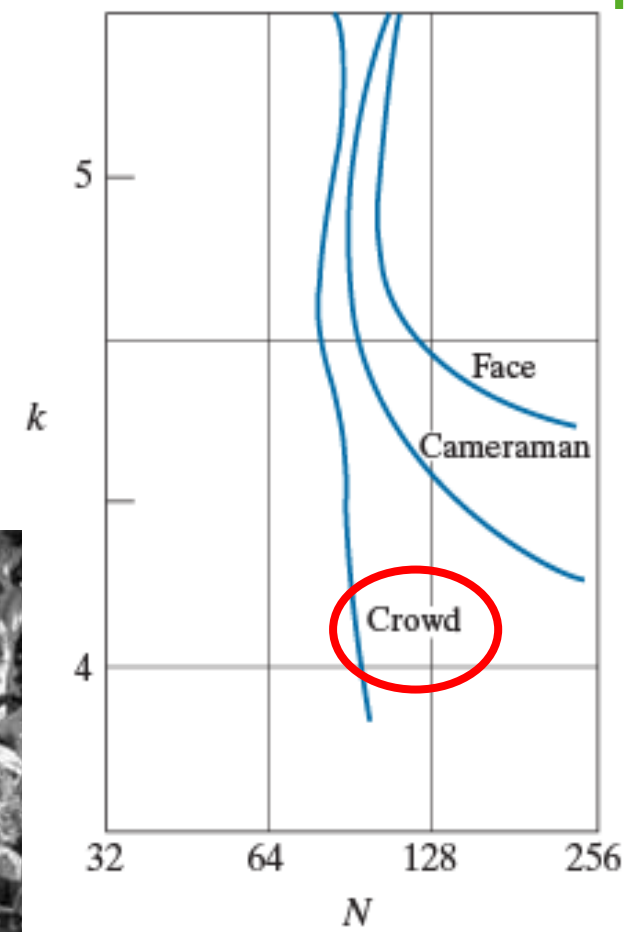
To Decide Spatial and Intensity Resolution: Isopreference Curve

FIGURE 2.26 Representative isopreference curves for the three types of images in Fig. 2.25.



Intensity Resolution: k

a b c

FIGURE 2.25 (a) Image with a low level of detail. (b) Image with a medium level of detail. (c) Image with a relatively large amount of detail. (Image (b) courtesy of the Massachusetts Institute of Technology.)

Image Resolution: N*N

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Image Interpolation

- **Zooming: increasing the number of pixels in an image so that the image appears larger**
  - Nearest neighbor interpolation
    - For example: pixel replication--to repeat rows and columns of an image
  - Bilinear interpolation
    - $v(x,y)=ax+by+cxy+d$
  - Bicubic interpolation

$$p(x,y)=\sum_{i=0}^{3}\sum_{j=0}^{3}a_{ij}x^i y^j.$$

- **Image shrinking: subsampling**

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Image Interpolation

Nearest neighbor

Bilinear

Bicubic

FIGURE 2.27 (a) Image reduced to 72 dpi and zoomed back to its original 930 dpi using nearest neighbor interpolation. This figure is the same as Fig. 2.23(d). (b) Image reduced to 72 dpi and zoomed using bilinear interpolation. (c) Same as (b) but using bicubic interpolation.

150

300

Image Fundamentals
Neighbors of Pixel

- **Neighbors of a pixel**
  - $N_4(p)$ 4-neighbors
    - the set of horizontal and vertical neighbors
  - $N_D(p)$ diagonal neighbors
    - the set of 4 diagonal neighbors
  - $N_8(p)$ 8-neighbors
    - union of 4-neighbors and diagonal neighbors

	O	
O	X	O
	O	

O	O	O
O	X	O
O	O	O

O		O
	X	
O		O

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Image Fundamentals
Adjacency

- **Adjacency:**
  - Two pixels that are neighbors and have the same grey-level (or some other specified similarity criterion) are adjacent
  - Pixels can be 4-adjacent, diagonally adjacent, 8-adjacent, or m-adjacent.
- ***m*-adjacency (mixed adjacency):**
  - Two pixels *p* and *q* of the same value (or specified similarity) are *m*-adjacent if either
    - (i) *q* and *p* are 4-adjacent, or
    - (ii) *p* and *q* are diagonally adjacent and do not have any common 4-adjacent neighbors.
    - They cannot be both (i) and (ii).

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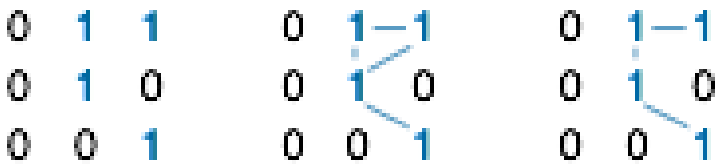
Image Fundamentals
Adjacency



a b c d e f

FIGURE 2.28 (a) An arrangement of pixels. (b) Pixels that are 8-adjacent (adjacency is shown by dashed lines). (c) *m*-adjacency. (d) Two regions (of 1's) that are 8-adjacent. (e) The circled point is on the boundary of the 1-valued pixels only if 8-adjacency between the region and background is used. (f) The inner boundary of the 1-valued region does not form a closed path, but its outer boundary does.

- ***m*-adjacency (mixed adjacency):**
  - Two pixels *p* and *q* of the same value (or specified similarity) are *m*-adjacent if either
    - (i) *q* and *p* are 4-adjacent, or
    - (ii) *p* and *q* are diagonally adjacent and do not have any common 4-adjacent neighbors.
    - They cannot be both (i) and (ii).

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Image Fundamentals
Relationships Between Pixels

- **Path**
  - $(x_0,y_0), (x_1,y_1),\ldots, (x_n,y_n)$; where (x_{i-1},y_{i-1}) and (x_i,y_i) are adjacent closed path if $(x_0,y_0) = (x_n,y_n)$
- **Connectivity in a subset *S* of an image**
  - Two pixels are connected if there is a path between them that lies completely within *S*.
- **Connected component of *S*:**
  - The set of all pixels in *S* that are connected to a given pixel in *S*.
- **Connected Set** if only one connected component
- **Region of an image:**
  - If the subset R is a connected set
- **Regions are adjacent:**
  - If their union forms a connected set

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Image Fundamentals
Relationships Between Pixels

- **Boundary/Contour (Inner Border)**
  - The set of points that are adjacent to the points in the complement of R
- **Outer Border**
  - The corresponding boundary of the background

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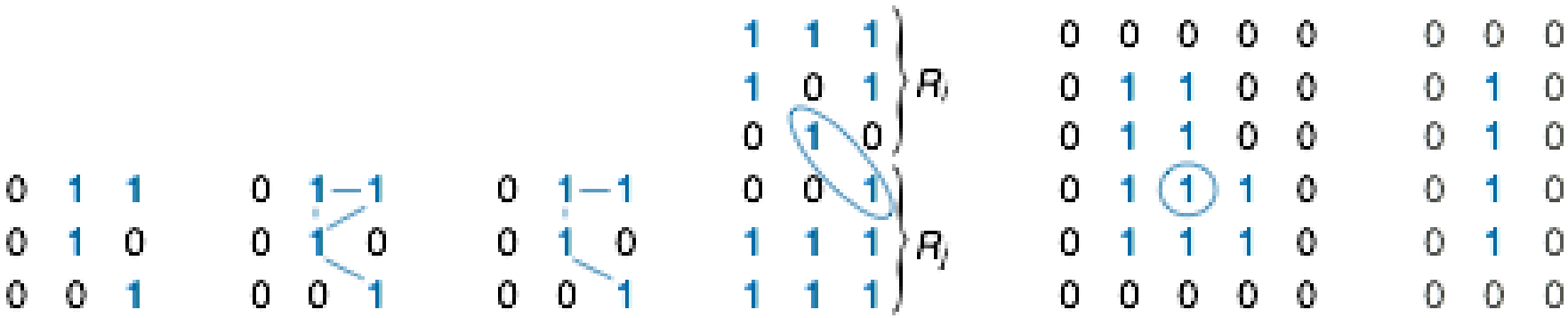
Image Fundamentals
Adjacency



a b c d e f

FIGURE 2.28 (a) An arrangement of pixels. (b) Pixels that are 8-adjacent (adjacency is shown by dashed lines). (c) *m*-adjacency. (d) Two regions (of 1's) that are 8-adjacent. (e) The circled point is on the boundary of the 1-valued pixels only if 8-adjacency between the region and background is used. (f) The inner boundary of the 1-valued region does not form a closed path, but its outer boundary does.

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❖ Distance measures

- Distance function: a function of two points, *p* and *q*, in space that satisfies three criteria

$$(a)\ D(p,q) \geq 0$$

$$(b)\ D(p,q) = D(q,p),\ \text{and}$$

$$(c)\ D(p,z) \leq D(p,q) + D(q,z)$$

Euclid Ευκλείδης 戰國

Pythagoras Πυθαγόρας 春秋

- The Euclidean distance $D_e(p, q)$

$$D_e(p,q) = \sqrt{(x-s)^2 + (y-t)^2}$$

- The city-block (Manhattan) distance $D_4(p, q)$

$$D_4(p,q) = |x-s| + |y-t|$$

- The chessboard distance $D_8(p, q)$

$$D_8(p,q) = \max(|x-s|, |y-t|)$$

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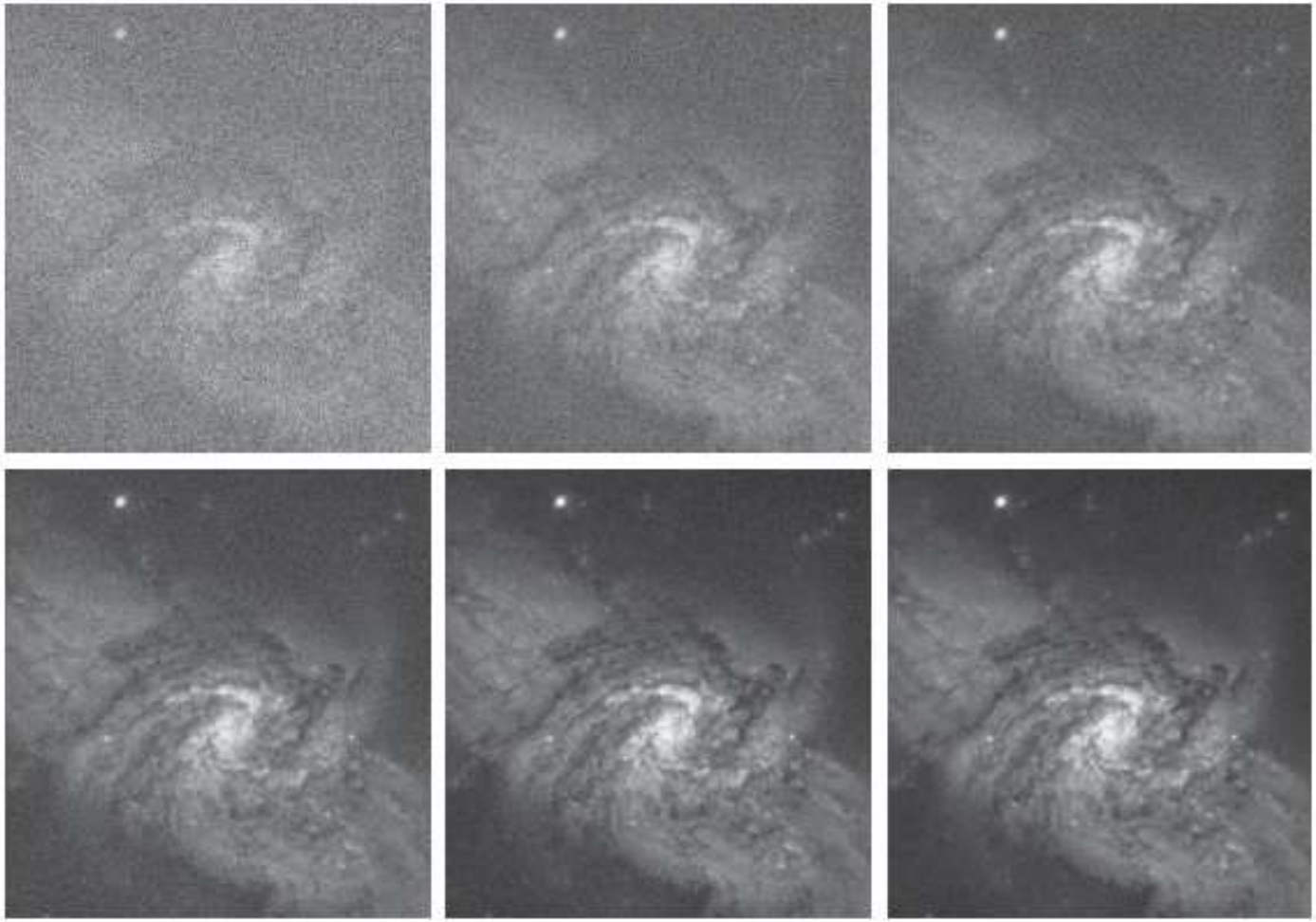
Image Fundamentals

Arithmetic Operations

❖ Image Averaging



a b c
d e f

FIGURE 2.29 (a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)-(f) Result of averaging 5, 10, 20, 50, and 1,00 noisy images, respectively. All images are of size 566 × 598 pixels, and all were scaled so that their intensities would span the full [0, 255] intensity scale. (Original image courtesy of NASA.)

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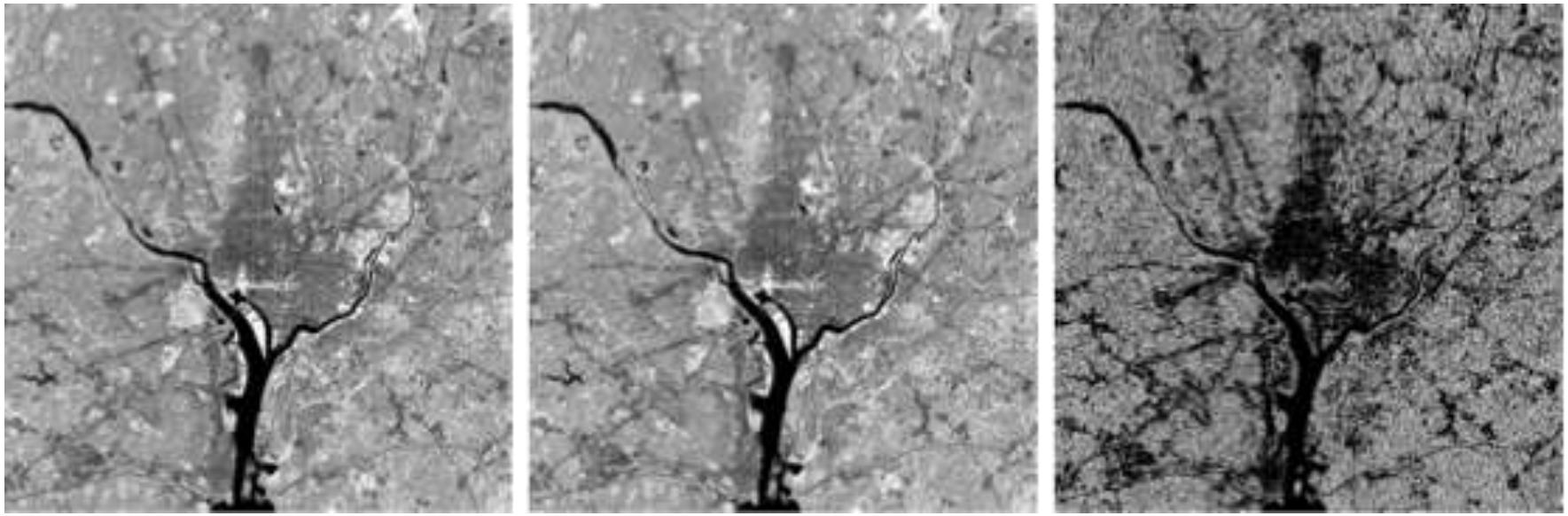
Image Fundamentals
Arithmetic Operations

❖ Image Subtraction



a b c

FIGURE 2.30 (a) Infrared image of the Washington, D.C. area. (b) Image resulting from setting to zero the least significant bit of every pixel in (a). (c) Difference of the two images, scaled to the range [0, 255] for clarity. (Original image courtesy of NASA.)

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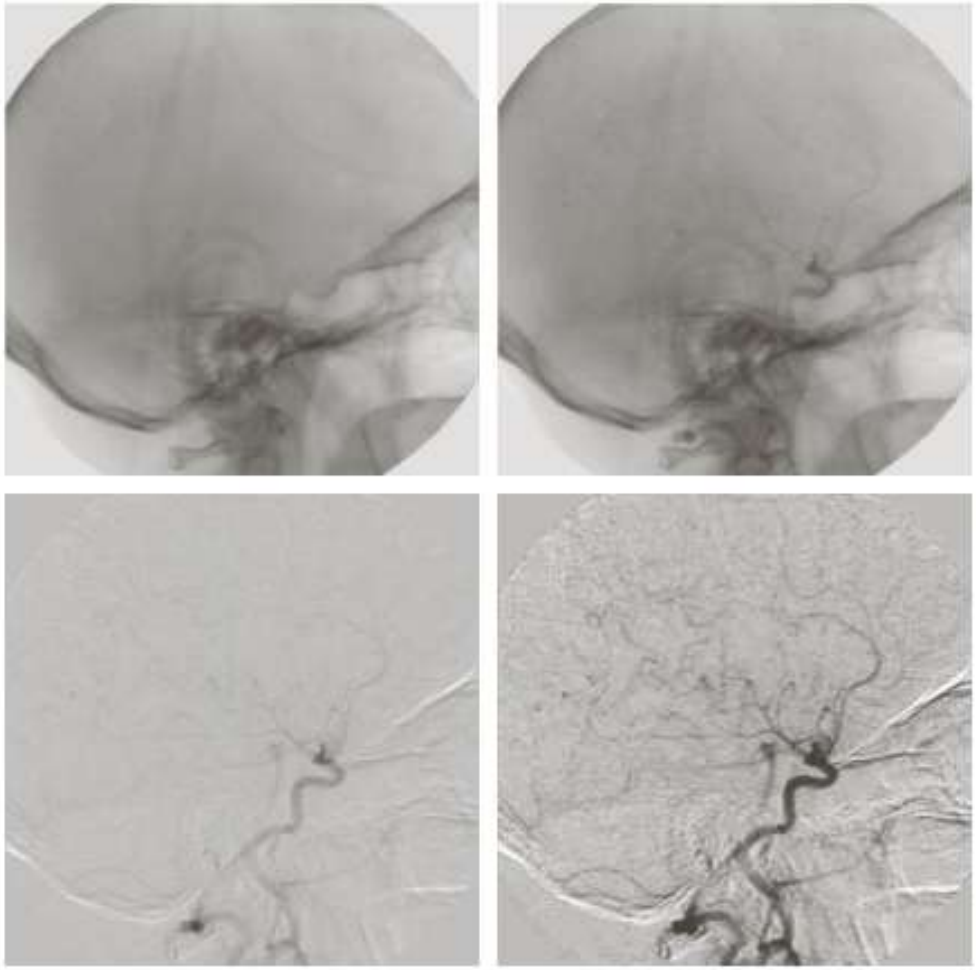
Image Fundamentals

Arithmetic Operations

❖ Image Subtraction



Contrast enhancement

a b
c d

FIGURE 2.32 Digital subtraction angiography. (a) Mask image. (b) A live image. (c) Difference between (a) and (b). (d) Enhanced difference image. (Figures (a) and (b) courtesy of the Image Sciences Institute, University Medical Center, Utrecht, The Netherlands.)

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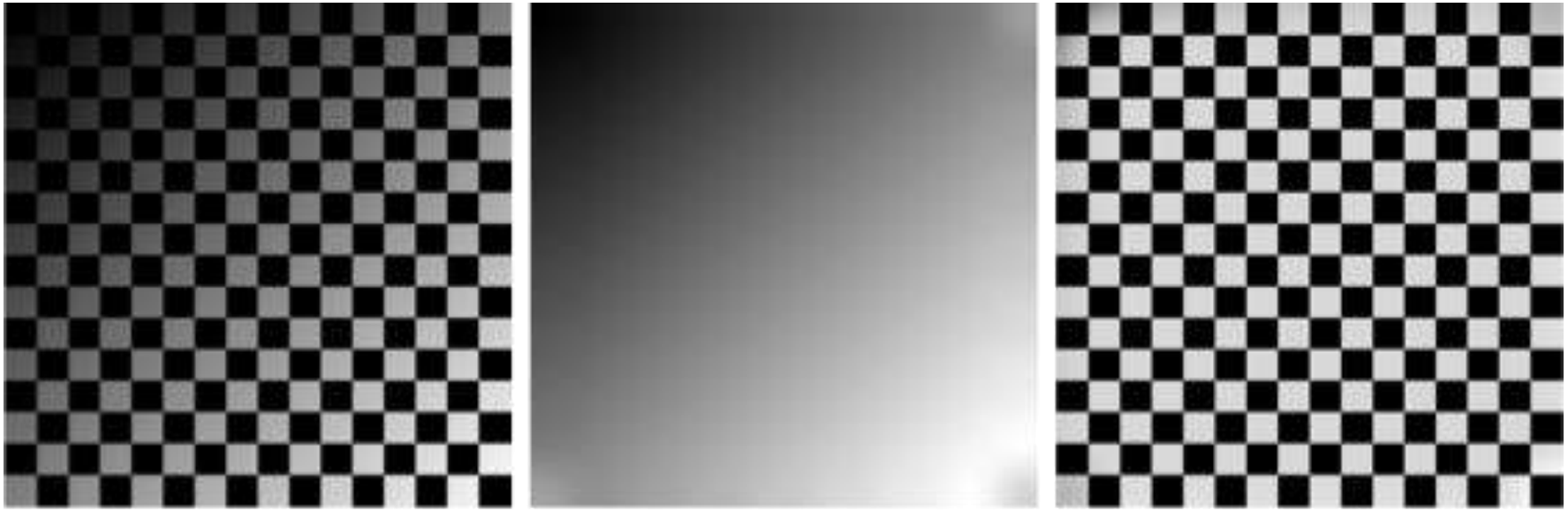
Image Fundamentals
Arithmetic Operations

❖ Shading correction



a b c

FIGURE 2.33 Shading correction. (a) Shaded test pattern. (b) Estimated shading pattern. (c) Product of (a) by the reciprocal of (b). (See Section 3.5 for a discussion of how (b) was estimated.)

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Image Fundamentals

Arithmetic Operations

❖ Image Multiplication

a b c

FIGURE 2.34 (a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).

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❖ Set Operations

TABLE 2.1 Some important set operations and relationships.

Description	Expressions
Operations between the sample space and null sets	$\Omega^c = \varnothing;\ \varnothing^c = \Omega;\ \Omega \cup \varnothing = \Omega;\ \Omega \cap \varnothing = \varnothing$
Union and intersection with the null and sample space sets	$A \cup \varnothing = A;\ A \cap \varnothing = \varnothing;\ A \cup \Omega = \Omega;\ A \cap \Omega = A$
Union and intersection of a set with itself	$A \cup A = A;\ A \cap A = A$
Union and intersection of a set with its complement	$A \cup A^c = \Omega;\ A \cap A^c = \varnothing$
Commutative laws	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributive laws	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$
DeMorgan's laws	$(A \cup B)^c = A^c \cap B^c$ $(A \cap B)^c = A^c \cup B^c$

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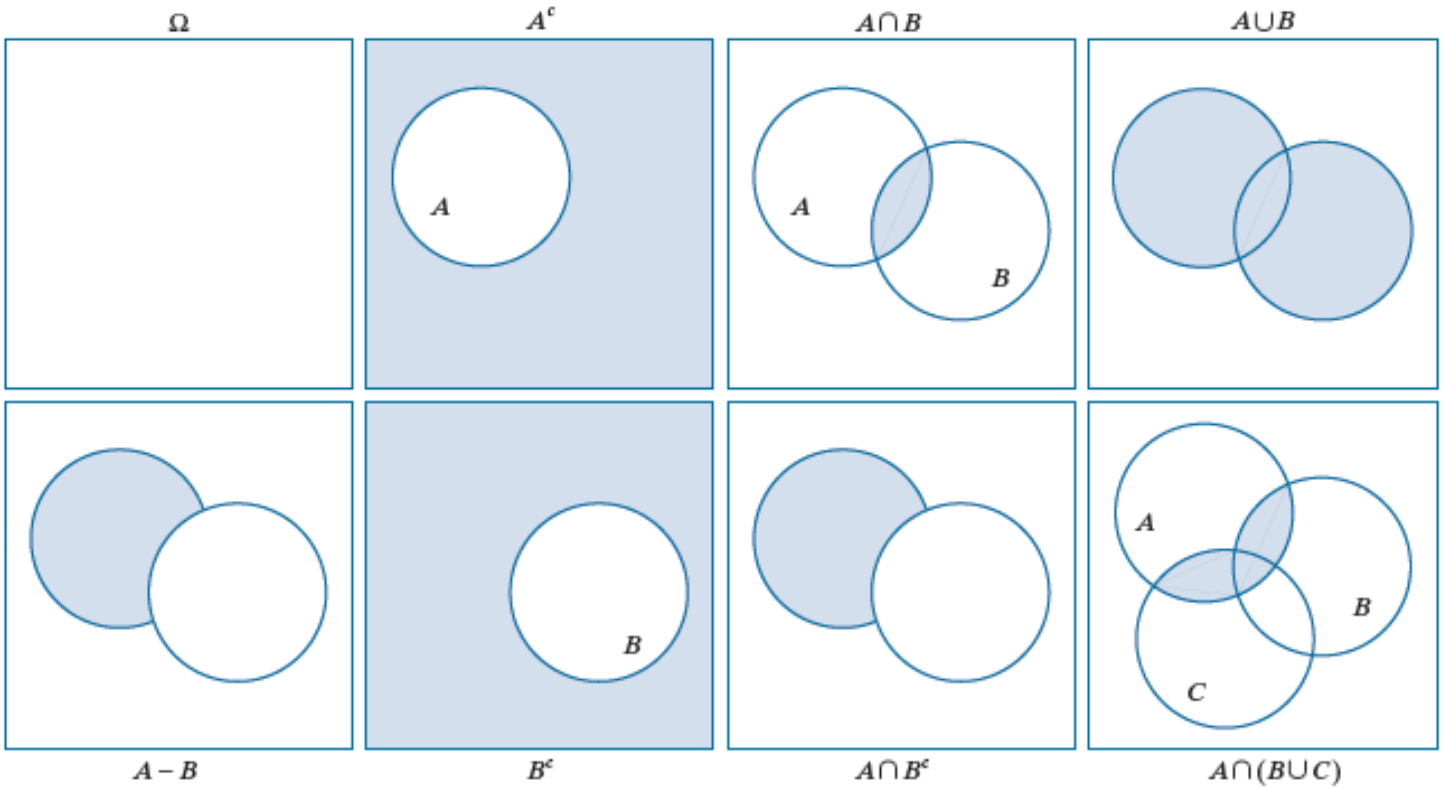
Image Fundamentals

Set and Logical Operations

Set Operations



a b c d
e f g h

FIGURE 2.35 Venn diagrams corresponding to some of the set operations in Table 2.1. The results of the operations, such as A^c, are shown shaded. Figures (e) and (g) are the same, proving via Venn diagrams that $A - B = A \cap B^c$ [see Eq. (2-40)].

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Image Fundamentals

Set and Logical Operations

Logical Operations

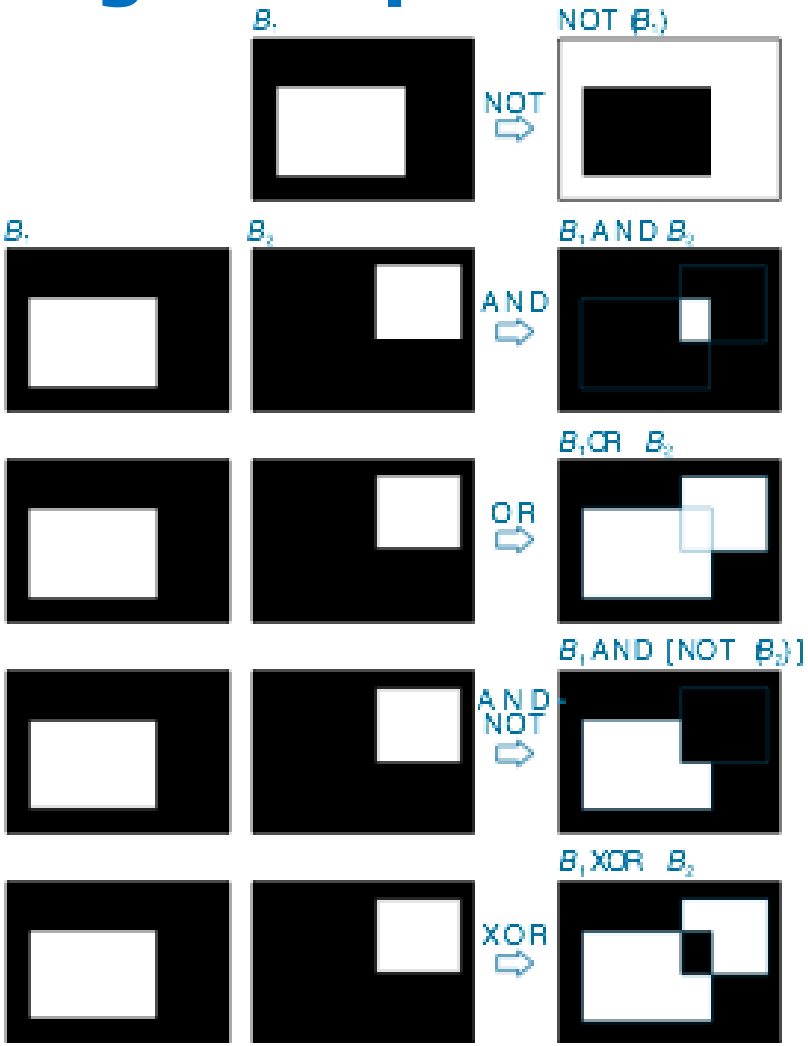

FIGURE 2.37 Illustration of logical operations involving foreground (white) pixels. Black represents binary 0's and white binary 1's. The dashed lines are shown for reference only. They are not part of the result.

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❖ Single-Pixel Operations

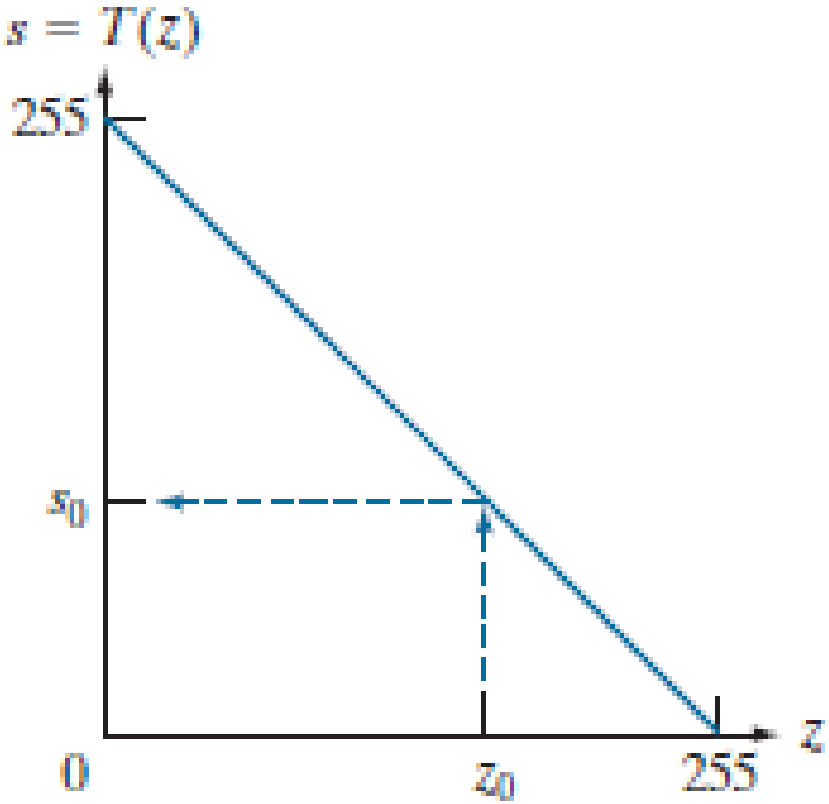

FIGURE 2.38 Intensity transformation function used to obtain the digital equivalent of photographic negative of an 8-bit image..

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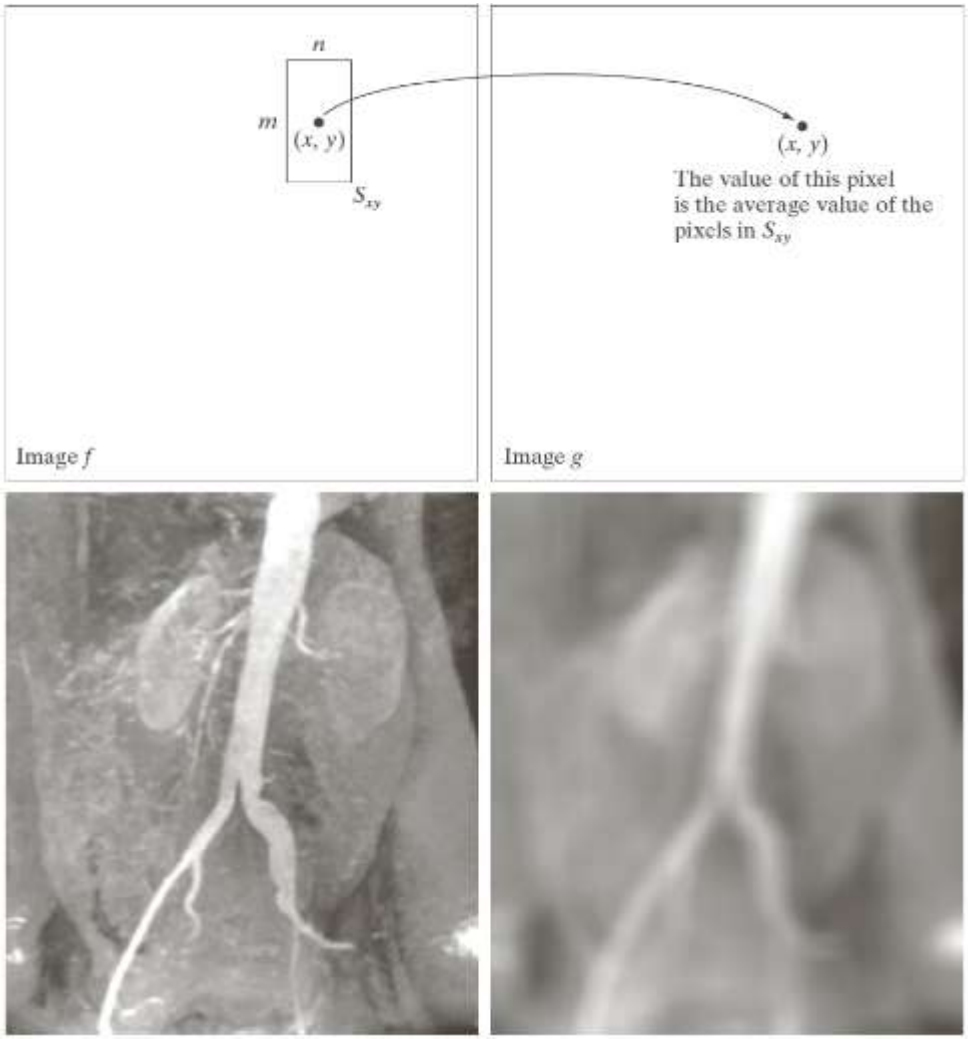
Image Fundamentals

Spatial Operations

❖ Neighborhood Operations



a b
c d

FIGURE 2.39 Local averaging using neighborhood processing. The procedure is illustrated in (a) and (b) for a rectangular neighborhood. (c) An aortic angiogram (see Section 1.3). (d) The result of using Eq. (2-43) with $m = n = 41$. The images are of size 790×686 pixels. (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

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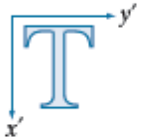
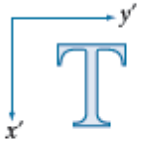
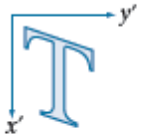
Geometric Spatial Transformations

TABLE 2.3 Affine transformations based on Eq. (2-45).

Transformation Name	Affine Matrix, A	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
Scaling/Reflection (For reflection, set one scaling factor to −1 and the other to 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	
Rotation (about the origin)	$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x\cos\theta - y\sin\theta$ $y' = x\sin\theta + y\cos\theta$	
Translation	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + t_x$ $y' = y + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + s_v y$ $y' = y$	
Shear (horizontal)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = s_h x + y$	

$[x'\ y'\ 1]^T = A[x\ y\ 1]^T$

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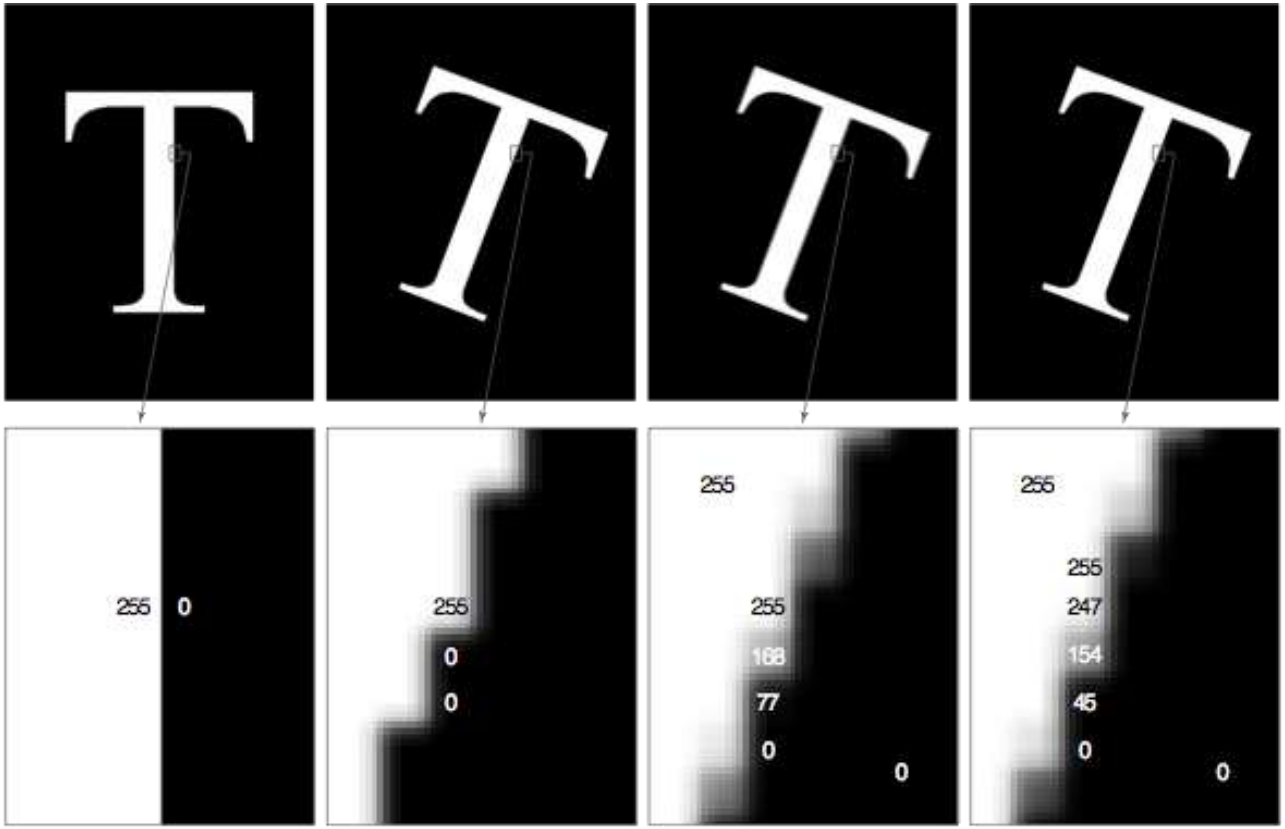
Image Fundamentals

Spatial Operations

❖ Geometric Spatial Transformations



a b c d
e f g h

FIGURE 2.40 (a) A 541 × 421 image of the letter T. (b) Image rotated −21° using nearest-neighbor interpolation for intensity assignments. (c) Image rotated −21° using bilinear interpolation. (d) Image rotated −21° using bicubic interpolation. (e)-(h) Zoomed sections (each square is one pixel, and the numbers shown are intensity values).

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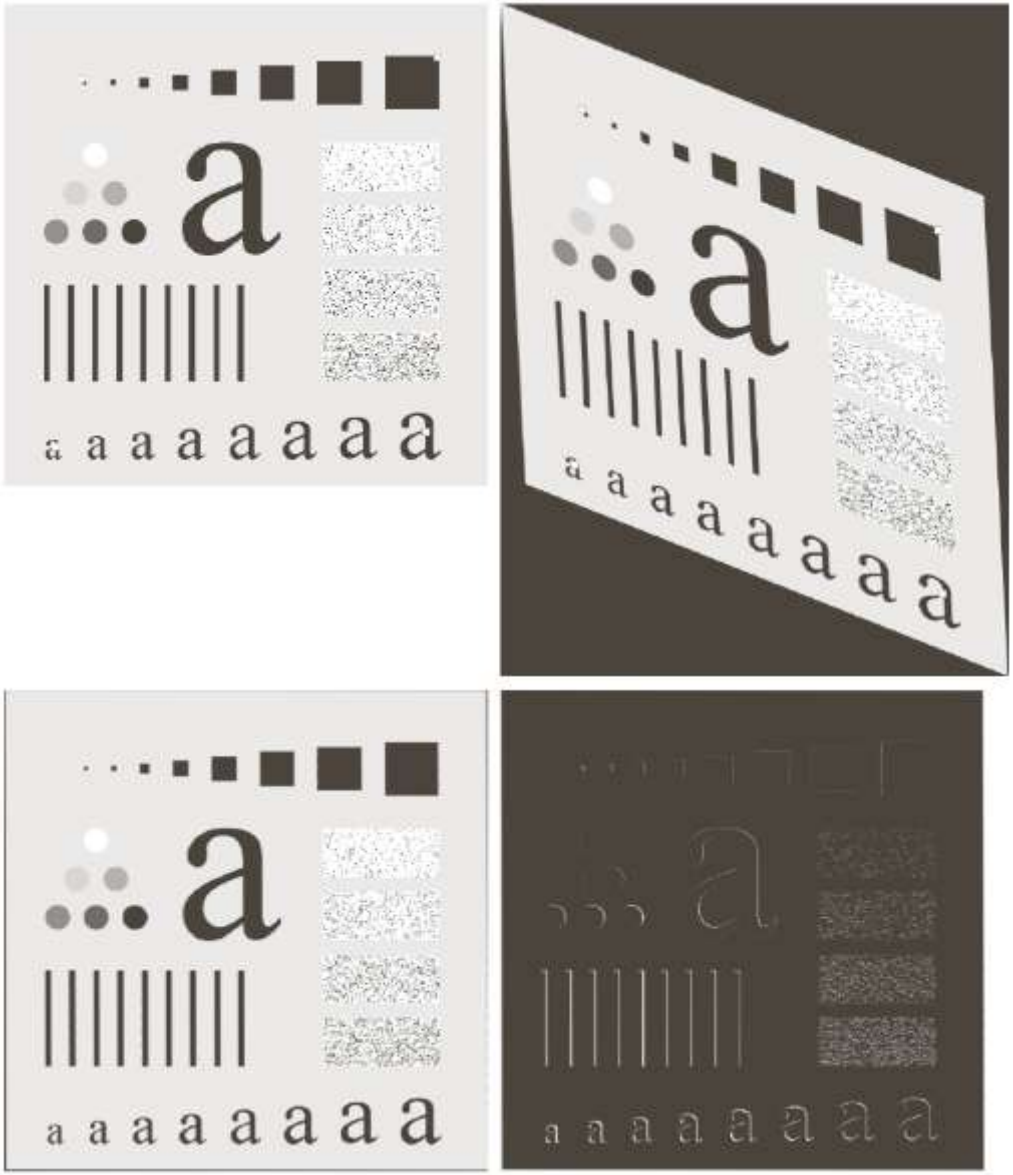
Image Fundamentals
Spatial Operations

❖ Registration

Bilinear Approximation

$$x=c_1v+c_2w+c_3vw+c_4$$
$$y=c_5v+c_6w+c_7vw+c_8$$



a b
c d

FIGURE 2.42 Image registration. (a) Reference image. (b) Input (geometrically distorted image). Corresponding tie points are shown as small white squares near the corners. (c) Registered (output) image (note the errors in the border). (d) Difference between (a) and (c), showing more registration errors.

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